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IONIC EQUILIBRIUM

Types of electrolytes
- Strong Electrolyte
- Weak Electrolyte
- Non- electrolyte.

1. Strong Electrolyte :
Which ionise almost completely in water.
eg : SA → HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_4$ etc.

SB → LiOH, NaOH, KOH, RbOH, CsOH, $Ba(OH)_2$, $Sr(OH)_2$ etc.

Salts → NaCl, KCl, Na_2CO_3, $BaCl_2$ etc.

→ Solutions of such strong electrolytes are highly conducting.

2. Weak Electrolytes :
Ionise partially in water.
eg. WA : RCOOH, HCN, HOCN, H_2CO_3, H_3PO_4 etc.
WB : NH_3 or NH_4OH, RNH_2, N_2H_4 (diacidic Base)

→ low conducting solution.

3. Non Electrolyte :
Dissolve in water but do not ionise
eg : glucose, urea, ROH etc.

→ Almost non- conducting solution.

* Arrhenius theory of dissociation :-

$$AB(aq) \rightleftharpoons A^+(aq) + B^-(aq), \quad K_{diss} = \frac{[A^+][B^-]}{[AB]}$$

	$AB(aq)$	$A^+(aq)$	$B^-(aq)$
$t=0$	C	0	0
$t=t_{eq}$	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$\Rightarrow K_{diss} = \frac{C\alpha \times C\alpha}{C(1-\alpha)} = \frac{(C\alpha)^2}{C(1-\alpha)} \quad \text{---(1)}$$

α = degree of dissociation / ionisation of substance

$$= \frac{\text{No. of moles of substance dissociated}}{\text{No. of moles of substance initially taken}}$$

$$= \frac{\text{कितना टूटा}}{\text{कितना लिया था।}}$$

Imp Factors affecting 'α' :-

(1.) Nature of electrolyte :-
α is nearly 1 for strong electrolyte and much less than 1 for weak electrolyte.

(2.) Nature of solvent :-
Dielectric constant of solvent ↑, α ↑

eg. $D_{(H_2O)(l)} \approx 80$

$$NaCl_{(s)} \xrightarrow{H_2O(l)} Na^+(aq) + Cl^-(aq)$$

- Water is excellent solvent for ionic compounds and polar covalent molecules.

(3) Concentration of solution :

(Refer to next lectures i.e Ostwald's dilution law)

(4) Temperature :-

→ For endothermic dissolution of electrolyte,
T↑, α↑

(5) Common ion Effect :-

eg. $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

In this solution, HCl(g) is passed / supplied

$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

→ When a common ion concentration of a weak electrolyte is increased, α of weak electrolyte is suppressed. (Common ion Effect)

Special Case:

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$ ↓ 0.1 M, 1L Solution

1 L, [HCl] = 0.01 M is added.

$$[H^+]_f = \frac{0.1 \times 1 + 0.01 \times 1}{2} = \frac{0.11}{2} = 0.055\ M$$

α_{CH_3COOH} → increases

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* Ostwald's dilution law :- (Effect of conc. or dilution variation on 'α')

(Applicable for weak electrolytes)

$$\underset{c}{AB(aq)} \rightleftharpoons \underset{0}{A^+(aq)} + \underset{0}{B^-(aq)} \quad , K_{diss.} = \frac{c\alpha^2}{1-\alpha} \quad -(i)$$

At eqm. $\quad c(1-\alpha) \qquad c\alpha \qquad c\alpha$

Conc. (c) = $\frac{n}{V}$

Concentration. ↑ ; Dilution. ↑

$$c \propto \frac{1}{\text{Dilution}}$$

For a weak electrolyte at moderate concentration, $(1-\alpha) \approx 1$.

From eq (i)

$$K_{diss.} = \frac{c\alpha^2}{1}$$

$$\alpha = \sqrt{\frac{K_{diss.} \times 1}{c}}$$

$$\alpha \cdot \propto \sqrt{K_{diss.} \times V}$$

$\therefore$

$$V\uparrow (\text{or } c\downarrow) \rightarrow \alpha\uparrow$$

From eq (1)

$$K_{diss.} = \frac{c\alpha^2}{1-\alpha} \Rightarrow c\alpha^2 + K_{diss} \times \alpha - K_{diss.} = 0$$

zero

$\therefore \alpha \to 1.$

$c \to 0$
(very dilute solution)

Important Conclusion :-

A weak electrolyte starts behaving like strong electrolyte when solution is extremely dilute ($\alpha \to 1$)

Such solution is known as infinitely dilute solution and molar conductivity of such solution does not further increase on further dilution.

* Acids and Bases :-

(1.) Arrhenius theory :-

Acid : which furnish H^+ in water

eg. HCl, H_2SO_4, HNO_3, $HClO_4$, CH_3COOH etc.

Base : which furnish OH^- in water.

eg: $NaOH$, $Ba(OH)_2$, $Ca(OH)_2$ etc.

(2) Bronsted - Lowry approach :-

Acid : Proton (H^+) donor

Base : Proton (H^+) acceptor

eg. $\underset{\text{acid}}{HCl(aq)} + \underset{\text{Base}}{H_2O(\ell)} \xrightleftharpoons[\text{very high}]{K_{eq} \text{ is}} H_3O^+ + Cl^-$

(3) Lewis acid - base theory :-

Lone Pair Donor : Lewis Base
Lone Pair Acceptor : Lewis Acid

eg. • $H_3N:$ + H^+ ⟶ NH_4^+
(H_3N: L-B, H^+: L-A)

• $H_3N:$ + BF_3 ⟶ $H_3N \rightarrow BF_3$
(LB) (LA)

* Bronsted lowry Approach (Detail)

Conjugate acid - base pair

eg. $HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$
acid-1, Base-1, acid-2, Base-2

Conjugate acid - base pair
↳ which differ by one proton (H^+)

eg. • $HCl - Cl^-$

$NH_4^+ \leftarrow NH_3 \rightarrow NH_2^-$

• $HCOOH_2^+ \leftarrow HCOOH \rightarrow HCOO^-$

• $NaH_2PO_4 + Na_2HPO_4$

• $H_3PO_4 + Na_3PO_4$ ⟶ conjugate acid - base pair.

Q. Which is/are not the conjugate acid-base pair

A. $HCl + NaCl$

B. $NH_3HPO_4 + Na_3PO_4$

C. $H_2CO_3 + Na_2CO_3$

D. $CH_3COOH + CH_3COOH_2^+$

* $HCl(aq) + H_2O(l) \xrightleftharpoons[High]{Keq} H_3O^+ + Cl^-$

Strong acid & strong base

Weak acid and weak base.

Note :-

(1.) An acid-base reaction proceeds in the direction which produces weaker acid and weaker base.

(2) For a conjugate acid-base pair,

$$K_a(\text{acid}) \times K_b(\text{conjugate base}) = K_w = 10^{-14} \text{ (at 25°C)}$$

Remember the ~~trick~~ —

⊙ Conjugate of very strong acid is a weak base

$HClO_4 \rightarrow ClO_4^-$

$HI \rightarrow I^-$

$HCl \rightarrow Cl^-$

very strong acid

Weak Bases.

⊙ Conjugate bases of very weak acids are strong bases.

eg. very weak acids: $H_2 \rightarrow H^-$, $NH_3 \rightarrow NH_2^-$, $CH_4 \rightarrow CH_3^-$ (very strong bases.)

$OH^- \rightarrow O^{2-}$

⊙ Conjugate base of weak acids is also weak.

weak base

eg. $CH_3COOH \rightarrow CH_3COO^-$
($K_a \approx 1.8 \times 10^{-5}$)

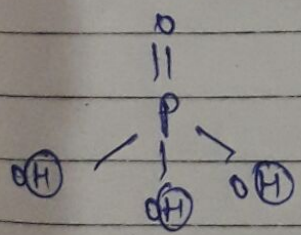
$$K_b(CH_3COO^-) = \frac{K_w}{K_a(CH_3COOH)} = \frac{10^{-14}}{1.8 \times 10^{-5}}$$

$$= 10^{-9}$$ hence, a very weak base

* For a moderately weak acid, conjugate base will not be a strong base but a weak base. Only for very weak acids, conjugate base will be a strong base

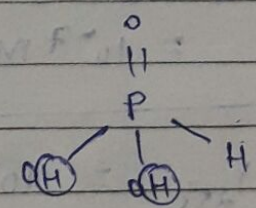
→ Basicity of acids →
No. of acidic H^+ per molecule of acid.

eg. H_3PO_4 (orthophosphoric acid)



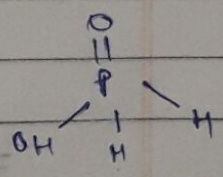
3-acidic 'H'
Basicity = 3
Tribasic acid (or) Tripotic acid

H_3PO_3 (Phosphorous acid)



2 acidic 'H'
Basicity = 2
Dipotic acid

H_3PO_2 (Hypophosphorous acid)



1 acidic 'H'
Basicity = 1
Monoprotic acid.

→ Acidity of a Base :
Ability to furnish no. of OH^- or to accept no. of H^+

eg. NaOH, KOH → Monoacidic Base.
$Ba(OH)_2$ → Diacidic Base.

* Self ionisation of water :-
(Properties of water)

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

$$K_{a(H_2O)} = K_{b(H_2O)} = \frac{[H^+][OH^-]}{[H_2O]}$$

$$[H_2O(l)] = \frac{1000}{18} \text{ mol/lit} = 55.5 \text{ M}$$

$$K_{a(H_2O)} \times [H_2O] = [H^+][OH^-]$$

↓

K_w (ionic product of water)

$$\Rightarrow K_w = [H^+][OH^-]$$

At 25°C, $[H^+] = [OH^-]$ in pure water (distill water)
$= 10^{-7}$ M

$$\Rightarrow \boxed{(K_w)_{25^\circ C} = 10^{-14} \text{ M}^2}$$

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$2NH_3(l) \rightleftharpoons NH_4^+ + NH_2^-$

$2BrF_3(l) \rightleftharpoons BrF_2^+ + BrF_4^-$

$2HCOOH(l) \rightleftharpoons HCOO^- + HCOOH_2^+$

Note :

$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$, Endothermic process.

$T\uparrow$, $\alpha_{H_2O}\uparrow$, $[H^+]\uparrow$, $[OH]\uparrow$, $KW\uparrow$

For aqueous solutions.

If $[H^+] > [OH^-]$, acidic
If $[H^+] < [OH^-]$, basic or alkaline
If $[H^+] = [OH^-]$, neutral solution.

} Applicable at Any temperature But At 25°C

If $[H^+] > 10^{-7}$ M, acidic
$[H^+] < 10^{-7}$ M, basic
$[H^+] = 10^{-7}$ M, neutral

* PH Scale :

(devised by P^H = — log $[H^+]$ total sorension)

$$p^{OH} = -\log [OH^-]$$

$$p^{Ka} = -\log Ka$$

$$p^{Kb} = -\log Kb$$

$$p^{Kw} = -\log Kw$$

For any aqueous solution,

$$[H^+][OH^-] = K_w$$

$$\log [H^+] + \log [OH^-] = \log K_w$$

$$-\log [H^+] - \log [OH^-] = -\log K_w$$

$$\boxed{p^H + p^{OH} = p^{Kw}}$$

Applicable for —

(i) Pure water

(ii) Any aqueous solution.

at any temperature

For pure water

$$[H^+] = 10^{-7} \text{ at } 25^\circ C$$

$$p^H = 7 = p^{OH}, \quad \boxed{p^{Kw} = 14} \text{ at } 25^\circ C$$

Nature of Solution :

At 25°C, If $p^H < 7$, acid.

$p^H > 7$, alkaline.

$p^H = 7$, neutral.

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* For pure water, $T\uparrow$, $\alpha_{H_2O}\uparrow$, $[H^+]\uparrow$ $pH\downarrow$

$[OH^-]\uparrow$ $p^{OH}\downarrow$, $K_w\uparrow$, $p^{Kw}\downarrow$

Q. At 60°C, K_w of water is 20 time that at 25°C An aqueous solution at 60°C is found to have $p^H = 6.8$. Comment whether the solution is acidic or basic or neutral.

$$(K_w)_{60^\circ C} = 20 \times 10^{-14}$$

$$[H^+][OH^-] = 20 \times 10^{-14} \quad \{\text{for pure water}\}$$

$$[H^+] = [OH^-] \Rightarrow [H^+]^2 = 20 \times 10^{-14}$$

$$[H^+]_{\substack{60^\circ C \\ (H_2O)}} = \sqrt{20} \times 10^{-7} \text{ M}$$

$$(p^H)_{60^\circ C, H_2O} = 7 - \log\sqrt{20}$$

$$= 7 - \frac{1}{2}(0.3 + 1)$$

$$= 7 - 0.65$$

$$= 6.35$$

Since p^H of given solution at 60°C > 6.35, so solution is alkaline / Basic.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

$1000/18$ — 0 — 0

$(1000/18 - x)$ — x — x

$$K_{a(H_2O)} = K_{b(H_2O)} = \frac{[H^+][OH^-]}{[H_2O]}$$

At 25°C, $[H^+] = 10^{-7}$ M

$$[H_2O] = \frac{1000}{18} \text{ M}$$

→ $$K_{a(H_2O)} = \frac{10^{-7} \times 10^{-7}}{\left(\frac{1000}{18}\right)} = 1.8 \times 10^{-16}$$ (Remember)

$x = [H^+] = 10^{-7}$

$$(\alpha_{H_2O})_{25°C} = \frac{x}{\left(\frac{1000}{18}\right)} = 18 \times 10^{-3} \times 10^{-7} = 1.8 \times 10^{-9} = \frac{1.8}{10^9}$$

→ α_{H_2O} is extremely low (even 2 molecules are not completely ionised out of 1 billion (10^9) molecules of water.

Therefore pure water is a poor conductor of electricity.

→ For a conjugate acid base pair,

$HA + A^-$

(conjugate base of HA)

$$K_{a(HA)} \times K_{b(A^-)} = K_w$$

Proof : $HA_{(aq)} \rightleftharpoons H^+_{(aq)} + A^-_{(aq)}$, $K_a = \frac{[H^+][A^-]}{[HA]}$

$A^-_{(aq)} + H_2O_{(l)} \rightleftharpoons HA_{(aq)} + OH^-_{(aq)}$, $K_b(A^-) = \frac{[HA][OH^-]}{[A^-]} \times \frac{[H^+]}{[H^+]}$

$\Rightarrow K_b(A^-) = \frac{K_W}{K_A(HA)}$

$$K_a(HA) \times K_b(A^-) = K_W$$

$$pK_a(HA) + pK_b(A^-) = pK_W$$

Applicable for every conjugate acid - base pair.

Eg. $H_2O \longrightarrow$ Base

$H_3O^+ \longrightarrow$ conjugate acid

$pK_a(H_3O^+) + pK_b(H_2O) = 14$

$K_b(H_2O) = 1.8 \times 10^{-16}$

$pK_{b(H_2O)} = 16 - \log 1.8$

$= 15.74$

$= pK_a(H_2O)$

$pK_a(H_3O^+) = 14 - 15.74$

$= -1.74$

→ Any acid stronger than H_3O^+ is a strong acid in water and almost completely ionised.

$HClO_4$, HBr, HCl, HNO_3, } strong acids

Eg. $H_2O \rightarrow$ acid, $OH^- \rightarrow$ it's conjugate base

$pK_a(H_2O) + pK_b(OH^-) = 14$

$pK_b(OH^-) = 14 - 15.74 = -1.74$

→ Any base stronger than OH^- is strong base and is completely ionised.

eg. NaOH, KOH, RbOH, $Ba(OH)_2$ etc.

- $O^{2-} + H_2O(l) \longrightarrow 2OH^-$
- $H^- + H_2O(l) \longrightarrow H_2\uparrow + OH^-$
- $CH_3^- + H_2O(l) \longrightarrow CH_4\uparrow + OH^-$
- $NH_2^- + H_2O(l) \longrightarrow NH_3\uparrow + OH^-$

Extremely Strong Bases.

* Salts Containing Amphiprotic anion :

Amphiprotic anion :
Anion which contains atleast one acidic hydrogen

eg. HCO_3^-, HS^-, $H_2PO_4^-$, HPO_4^{2-}, $H_2CO_4^{2-}$ etc.

↓ ↓ ↓

$NaHCO_3$ NaHS NaH_2PO_4

$$NaHCO_3(s) \xrightarrow{H_2O(l)} Na^+ + HCO_3^-(aq)$$

$$HCO_3^- + H_2O(l) \rightleftharpoons H_3O^+ + CO_3^{2-} \quad (\text{Acidic})$$
$$HCO_3^- + H_2O(l) \rightleftharpoons H_2CO_3 + OH^- \quad (\text{Basic})$$

Ans. Q Which of the following Salts contain amphiprotic anion

(A) Na_2HPO_3

(B) NaH_2PO_2

(C) NaHS

(D) NaH_2PO_3

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(*) Levelling Effect :

$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

$HBr(aq) \longrightarrow H^+(aq) + Br^-(aq)$

$HI(aq) \longrightarrow H^+(aq) + I^-(aq)$

→ In water these (above) acids are equally strong. ~~There~~ Their acidic strength can't be compared in water, So water is called levelling solvent for these acids,

In other words, water has levelled these acids on same level of acidic strength. This is called levelling effect.

* To compare acidic strength of HCl, HBr, HI, use solvent which is a weaker base than water.
eg. RCOOH.

$HCl + RCOOH \rightleftharpoons RCOOH_2^+ + Cl^-$

↓

{ Differentiating solvent } Strong acids in water

$HClO_4 > HI > HBr > HCl > HNO_3 > H_3PO_4 > HCOOH > CH_3COOH > HCN$

* Calculation of PH : (in aqueous solution)

PH calculation for strong acids and strong bases :

HCl, HBr, HI, $HClO_4$, H_2SO_4 etc.

NaOH, KOH, $Ba(OH)_2$ etc.

Eg. HCl
C mol/lit

HCl	$\longrightarrow$	$H^+_{(aq)}$	+	$Cl^-_{(aq)}$
C		0		0
0		(C+x)		C

$$H_2O(l) \rightleftharpoons H^+ + OH^-(aq)$$

~~10^{-7} M~~ (in pure water at 25°C)

$(x+c)$ x

$$[H^+][OH^-] = K_w$$

$$x(x+c) = 10^{-14}$$

$$x^2 + cx - 10^{-14} = 0$$

$$x = \frac{-c + \sqrt{c^2 + 4 \times 10^{-14}}}{2}$$

Appropriate approximations :

1. If conc. of S.A $\geq 10^{-6}$ M, then $[H^+]$ from water can be ignored. , $(x \ll\ll c)$

$$[H^+] = c + x$$

$$pH = -\log c$$ Ans

2. If $10^{-8} M \leq C < 10^{-6} M$

$[H^+]$ from water can't be ignored.

$[H^+] = C + x$

So, Solve Quadratic and get x.

3. If $C < 10^{-8} M$

$[H^+]$ from S.A can be ignored.

In that case,

pH is nearly 7.

LECTURE 7 (new one was a re-upload having different questions, this one is as per old video)

Q Find pH in following aqueous solution :-

(i) 2×10^{-3} M $HCl(aq)$

(ii) 4×10^{-4} M $Ba(OH)_2(aq)$

(iii) 2×10^{-7} M $HBr(aq)$

Given –

$\log_{10} 2 = 0.3$

$\log_{10} 3 = 0.48$

Soln. (i) 2×10^{-3} M $HCl(aq)$

$[H^+] = 2 \times 10^{-3} M,$

$pH = -\log 2 \times 10^{-3}$

$= 3 - \log 2 = 2.7$

(ii) 4×10^{-4} M $Ba(OH)_2$

$[OH^-] = 2 \times 4 \times 10^{-4} = 8 \times 10^{-4} M$

$pOH = 4 - \log 8$

$= 4 - \log 2^3$

$= 4 - 3\log 2 = 3.1$

$\Rightarrow pH = 14 - 3.1$

$= 10.9$

(iii) 2×10^{-7} M $HBr(aq)$

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$
$$(x + 2\times10^{-7}) \quad x$$

$$[H^+][OH^-] = 10^{-14}$$
$$x(x + 2\times10^{-7}) = 10^{-14}$$

$$x^2 + 2\times10^{-7}x - 10^{-14} = 0$$

$$x = \frac{-2\times10^{-7} + \sqrt{4\times10^{-14} + 4\times10^{-14}}}{2}$$

$$\Rightarrow x = 0.4\times10^{-7} \text{ M}$$

$$[H^+]_{total} = x + 2\times10^{-7}$$
$$= 2.4\times10^{-7} \text{ M}$$
$$= 24\times10^{-8} \text{ M}$$

$$pH = 8 - \log 24$$
$$= 8 - \log 2^3 \times 3$$
$$= 8 - 0.9 - 0.48$$
$$= 8 - 1.38$$
$$= 6.62 \text{ Ans}$$

Q: Find pH of 10^{-7} M $Ba(OH)_2(aq)$

Soln: $H_2O(l) \rightleftharpoons H^+ + OH^-$

$\frac{1000}{18} - x \quad x \quad (x + 2\times10^{-7})$

$$x(x + 2\times10^{-7}) = 10^{-14}$$

$$x = 0.4\times10^{-7} \text{ M}$$

$[H^+] = x = 4\times10^{-8} \Rightarrow pH = 8 - 0.6 = 7.4$ Ans

* pH of mixture of Strong Acid and Strong Base :-

Q. (a) 2 Lit, 0.6M $HCl(aq)$ + 2 Lit, 0.2M $HBr(aq)$

(b) 100ml, 0.2M $H_2SO_4(aq)$ + 100ml, 0.1M $Ba(OH)_2(aq)$

Baryta water

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Find pH of resulting solution —

soln (a) $[H^+]_f = \frac{2 \times 0.6 \times 1 + 2 \times 0.2 \times 1}{2+2}$

$= \frac{1.2 + 0.4}{4} = 0.4\ M$

$pH = -\log 4 \times 10^{-1}$

$= 1 - 0.6 = 0.4$ Ans.

(b)

$H_2SO_4(aq) + Ba(OH)_2 \longrightarrow BaSO_4\downarrow + 2H_2O$

20 mmol — 10 mmol

−10 — (L.R)

= 10 mmol

$[H^+] = \frac{10 \times 2}{200} = 0.1\ M$

$pH = -\log 10^{-1}$

$= 1$ Ans.

Q: Find pH, 100 ml 0.1M H_2SO_4 + 100 ml, 0.1M KOH

Soln: $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O(l)$

$100 \times 0.1 = 10$ mmol — $100 \times 0.1 = 10$ mmol

− 5 mol — (L.R)

= 5 mmol

$[H^+] = \frac{5}{200} \times 2 = \frac{1}{20}\ M = 5 \times 10^{-2}\ M$.

$pH = 2 - \log 5 = 1.3$ Ans

LECTURE 7 ENDS

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* pH Calculation of monoprotic weak acid or monoacidic weak base solution :-

eg. Monoprotic / Monobasic weak acid

$HCOOH$, CH_3COOH, C_6H_5COOH, HCN etc.

eg. Monoacidic Bases :

NH_4OH, $C_6H_5NH_3OH$ etc.

Weak acid	Weak Base
↓	↓
$HA \rightarrow RCOOH$	$BOH \rightarrow NH_4OH$
$H_2A \rightarrow H_2CO_3$	$B(OH)_2 \rightarrow N_2H_4$
$H_3A \rightarrow H_3PO_4$	

eg. $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

c	0	0
$c(1-\alpha)$	$c\alpha$	$c\alpha$

$$K_a = \frac{c\alpha^2}{1-\alpha}$$

Appropriate approximations :-

Case I- If concentration of weak acid, $HA \geq 10^{-2}$ M from ostwald's dilution law, $\alpha << 1$

$\therefore (1-\alpha) \approx 1$

$$K_a = c\alpha^2$$

$$\alpha = \sqrt{\frac{K_a}{c}} \Rightarrow [H^+] = c\alpha = \sqrt{K_a \times c}$$

$$-\log[H^+] = \frac{1}{2}\left(pK_a - \log c\right)$$

$$\Rightarrow \boxed{pH = \frac{1}{2}\left(pK_a - \log c\right)} \rightarrow \text{Applicable when } \alpha \leq 0.1$$

* Weak monoacidic base, NH_4OH or aq. NH_3

$$\boxed{pOH = \frac{1}{2}\left(pK_b - \log c\right)} \rightarrow \text{applicable when } \alpha \leq 0.1$$

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Case II - When conc. of WA $10^{-6}\,M \leq c < 10^{-2}\,M$
(New solution is dilute),
From ostwald's dilution law, now α can't be ignored w.r.t 1.
Solve quadratic eq (1) i.e $K_a = \frac{c\alpha^2}{1-\alpha}$

**** Note :

(i) Don't go to case-I, or case-II directly
→ first calculate approximate α as

$$\alpha_{app.} = \sqrt{\frac{K_a}{c}} \quad \left(\text{or } \alpha = \sqrt{\frac{K_b}{c}}\right)$$

→ If this $\alpha \leq 0.1$, then go to case-I or else go to case-II.

** Case-I and case-II are designed based only on 'c', however, α depends on both K_a (or K_b) as well as 'c'.

$$K_a = \frac{c\alpha^2}{1-\alpha}$$

(*) Case-III

If solution of weak acid is very dilute
i.e $10^{-8}\,M \leq c < 10^{-6}\,M$
Then treat weak acid as strong Acid and assume complete ionisation. Also, consider effect of water.

HA	⟶	H^+	+	A^-
c		0		0
0		c		c

$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$

$(x+c) \qquad x$

$(x+c) \times x = 10^{-14}$

Find x. $[H^+] = c+x$

Now you can calculate pH of solution.

Advanced.

(*) Case - IV

When K_a or K_b of WA (or WB) is very small i.e when $K_a \times C \approx 10^{-13}, 10^{-14}, 10^{-15}$ etc.
In such case also, effect of water should be consider infact treat the problem as mixture of two weak acids.

Q. 0.2 M aqueous NH_3 solution, $K_a(NH_4^+) = 10^{-9}$. Find pH of this solution.

Soln $NH_4OH(aq) \rightleftharpoons NH_4^+ + OH^-$

$$\alpha_{approx.} = \sqrt{\frac{K_b}{C}} \simeq \sqrt{\frac{10^{-5}}{0.2}} = \sqrt{50 \times 10^{-6}} = 7.1 \times 10^{-3} \ll 1$$

$$K_b(NH_3) \times K_a(NH_4^+) = K_w$$

$$K_b = \frac{10^{-14}}{10^{-9}}$$

$$K_b = 10^{-5}$$

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$$pOH = \frac{1}{2}(pK_b - \log c)$$

$$= \frac{1}{2}(5 - \log 2\times10^{-1})$$

$$= \frac{1}{2}(5 + 1 - 0.3) = \frac{5.7}{2} = 2.85$$

$$pH = 14 - pOH$$

$$= 14 - 2.85$$

$$= 11.15 \quad \underline{\text{Ans.}}$$

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$

$0.2 \quad 0 \quad 0, \quad K_b = 10^{-5} = \frac{x^2}{0.2-x}$

$0.2-x \quad x \quad x$

$\alpha_{NH_3} = \frac{x}{0.2}$

$= \frac{\sqrt{2}\times10^{-3}}{0.2}$

$= 5\times1.4\times10^{-3}$

$= 7\times10^{-3}$

Since $\alpha < 0.1$, so $x << 0.2$

$(0.2-x) \cong 0.2$

$\Rightarrow \frac{x^2}{0.2} = 10^{-5} \Rightarrow x = \sqrt{2\times10^{-6}}$

$= \sqrt{2}\times10^{-3}\ M$

$[OH^-] = x = \sqrt{2}\times10^{-3}$

$pOH = 3 - \frac{1}{2}\times0.3 = 2.85$

$pH = 11.15$ ✓

Q. 0.2 M aq. CH_3COOH ($K_a = 2\times10^{-5}$). Find.

(i) pH (ii) α_{CH_3COOH}. (iii) α_{H_2O}

Soln. $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$,

$0.2-x \quad x \quad x$

$$K_a = \frac{x^2}{0.2-x}$$

$$2\times10^{-5} = \frac{x^2}{2} \quad \{\because (x << 0.2)\}$$

$$\Rightarrow x = \sqrt{4\times10^{-6}}\ M = 2\times10^{-3}\ M$$

(i) $pH = 3 + \log 2$

$= 2.7$ Ans

(ii) $\alpha_{CH_3COOH} = \frac{x}{0.2} = \frac{2 \times 10^{-3}}{0.2} = 10^{-2}$

Imp (iii) $H_2O(l) \rightleftharpoons H^+_{(aq)} + OH^-_{(aq)}$

$\frac{1000}{18} - y$ $\quad$ 2×10^{-3} M $\quad$ y

$K_w = [H^+][OH^-]$

$10^{-14} = 2 \times 10^{-3} \times y$

$y = 0.5 \times 10^{-11}$

$= 5 \times 10^{-12}$ M

$\alpha_{H_2O} = \frac{y}{\left(\frac{1000}{18}\right)} = 5 \times 10^{-12} \times 18 \times 10^{-3}$

$= 90 \times 10^{-15}$

$= 9 \times 10^{-14}$

* $\alpha_{(H_2O)_{\text{Pure water}}} = 1.8 \times 10^{-9}$ at 25°C

Imp Q. An aqueous solution of HCl has pH = 6.5. Find concentration of HCl.

$[HCl] = c$

Soln $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$

$(x+c)$ $\quad$ x

$[H^+]_{total} = (x+c) = 10^{-6.5}$ M

$= \sqrt{10} \times 10^{-7}$ M.

$\sqrt{10} \times 10^{-7} \times x = 10^{-14}$

$x = \frac{1}{\sqrt{10}} \times 10^{-7}$ M.

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$$c = \sqrt{10} \times 10^{-7} - \frac{1}{\sqrt{10}} \times 10^{-7}$$

$$c = \frac{9}{\sqrt{10}} \times 10^{-7} \text{ M}$$

$$c = 2.8 \times 10^{-7} \text{ M}.$$

* **Note:**

Whenever pH of solution is more than 6 and less than 8, then effect of water must have been considered.

* pH of mixture of two weak acids or two weak base :

$$\underset{(c_1)-x}{HA(aq)} \rightleftharpoons \underset{(x+y)}{H^+(aq)} + \underset{x}{A^-(aq)}, \quad Ka_1 = \frac{x(x+y)}{c_1 - x} \quad \text{---(i)}$$

$$\underset{(c_2)-y}{HB(aq)} \rightleftharpoons \underset{(y+x)}{H^+(aq)} + \underset{y}{B^-(aq)}, \quad Ka_2 = \frac{y(x+y)}{c_2 - y} \quad \text{---(ii)}$$

Appropriate approximations :-

1. If $\alpha_1 \left(= \sqrt{\frac{Ka_1}{c_1}} \right) \leq 0.1$, $x \ll\ll c_1$
   $\Rightarrow (c_1 - x) \approx c_1$

If $\alpha_2 \left(= \sqrt{\frac{Ka_2}{c_2}} \right) \leq 0.1$, $y \ll\ll c_2$
$(c_2 - y) \approx c_2$

Now,

$$x(x+y) = Ka_1 C_1$$

$$y(x+y) = Ka_2 C_2$$

On adding $(x+y)^2 = Ka_1C_1 + Ka_2C_2$

$$[H^+] = (x+y) = \sqrt{Ka_1C_1 + Ka_2C_2}$$

$$pH = -\log\left(\sqrt{Ka_1C_1 + Ka_2C_2}\right)$$

$$[A^-] = x = \frac{Ka_1C_1}{\sqrt{Ka_1C_1 + Ka_2C_2}}$$

$$[B^-] = y = \frac{Ka_2C_2}{\sqrt{Ka_1C_1 + Ka_2C_2}}$$

* For a mixture of two weak bases.

$$[OH^-] = \sqrt{Kb_1C_1 + Kb_2C_2}$$

Q. 200 ml, 0.4M, HCOOH ($Ka = 5 \times 10^{-5}$)
+ 200ml, 0.1M CH_3COOH ($Ka = 2 \times 10^{-5}$)

↓

Resulting Solution ₹.

Find pH of resulting solution.

Sol.

$[HCOOH]_{final} = 0.2M$

$[CH_3COOH]_{final} = 0.5M$

$[H^+]_{total} = \sqrt{Ka_1c_1 + Ka_2c_2}$

$= \sqrt{5\times10^{-5}\times0.2 + 2\times10^{-5}\times0.5}$

$= \sqrt{20\times10^{-6}}$

$= \sqrt{20}\times10^{-3}\,M$

$pH = -\log(\sqrt{20}\times10^{-3})$

$= 3 - \frac{1}{2}(0.3+1)$

$= 3 - 0.65$

$= 2.35$ Ans.

Imp Q. Find pH of 0.01 M aq. weak acid HA solution.
$(Ka(HA) = 10^{-12}$

$[H^+] = \sqrt{Ka\times c} = \sqrt{10^{-12}\times10^{-2}} = 10^{-7}\,M$

Effect of water should be considered

$HA \rightleftharpoons H^+ + A^-$, $10^{-12} = \frac{x(x+y)}{0.01-x}$

$H_2O(l) \rightleftharpoons H^+ + OH^-$

$(0.01-x)$ $(x+y)$ (x)

$\left(\frac{1000}{18}-y\right)$ $(y+x)$ y ,

$1.8\times10^{-16} = \frac{y(x+y)}{\left(\frac{1000}{18}-y\right)}$

$x(x+y) = 10^{-14}$

$y(x+y) = 10^{-14}$

$[H^+] = (x+y) = \sqrt{2}\times10^{-7}\,M$

$$pH = 7 - \frac{1}{2} \times 0.3$$

$$= 6.85 \quad \text{Ans.}$$

(*) PH calculation of a mixture of W.A and S.A

$$\underset{c_1}{CH_3COOH} + \underset{c_2}{HCl}$$

$$\underset{c_1 - x}{CH_3COOH(aq)} \rightleftharpoons \underset{x}{CH_3COO^-} + \underset{(x+c_2)}{H^+}$$

$$K_a = \frac{x\{x + c_2\}}{c_1 - x} \quad \text{---} \quad (1)$$

$$x^2 + (c_2 + K_a)x - K_a c_1 = 0$$

$$x = \frac{-(c_2 + K_a) + \sqrt{(c_2 + K_a)^2 + 4K_a c_1}}{2}$$

Approximations :

(1) If $K_a \ll c_2$, $(K_a + c_2) \approx c_2$

$$x = \frac{-c_2 + \sqrt{c_2^2 + 4K_a c_1}}{2}$$

(2) If $4K_a c_1 \ll c_2^2 \Rightarrow x \approx 0$

$[H^+] = c_2$,

Now pH can be calculated.

Q. 1 lit. of 0.2 M formic acid solution is mixed with 1 lit. of 0.1 M HBr (aq) solution. Find pH of Resulting solution and also calculate degree of ionisation of HCOOH. [$K_a = 10^{-4}$] for HCOOH.

Soln. $[HCOOH]_{final} = 0.1\,M$

$[HBr]_{final} = 0.05\,M$

$HCOOH(aq) \rightleftharpoons HCOO^-_{(aq)} + H^+(aq)$

$0.1 - x \qquad x \qquad (x + 0.05)$

$[H^+] = x + 0.05$

$pH = -\log 5 \times 10^{-2}$

$= 2 - 0.7$

$\boxed{pH = 1.3}$

$10^{-4} = \dfrac{x(x + 0.05)}{0.1 - x}$

$x = \dfrac{10^{-4} \times 0.1}{0.05}$

$x = 2 \times 10^{-4}\,M.$

$\alpha_{HCOOH} = \dfrac{2 \times 10^{-4}}{0.1}$

$= 2 \times 10^{-3}$ Ans.

* pH of a polyprotic acid or a polyacidic Base.

H_2CO_3, H_2S, $H_2C_2O_4$, H_3PO_4

Diprotic acid — Triprotic acid

NH_4OH, $C_6H_5NH_2$, N_2H_4 (or $N_2H_6(OH)_2$)

monoacidic Base — Diacidic Base.

HA	BOH
H_2A	$B(OH)_2$
H_3A	$B(OH)_3$

eg. $H_2A\,(aq) \rightleftharpoons H^+ + HA^-$

$c-x \qquad (x+y) \quad x-y$

$HA^-(aq) \rightleftharpoons H^+(aq) + A^{2-}(aq)$

$x-y \qquad (y+x) \quad y$

$$K_{a_1} = \frac{(x+y)(x-y)}{c-x} \quad \text{—— (i)}$$

$$K_{a_2} = \frac{(x+y)\,y}{x-y} \quad \text{—— (ii)}$$

Approximations :

1. For polyprotic acids,

$K_{a_1} > K_{a_2} > K_{a_3} \ldots\ldots$

$\therefore \; y << x$

$$K_{a_1} = \frac{x^2}{c-x} \quad \text{—— (iii)}$$

$$K_{a_2} = y = [A^{2-}] \quad \text{—— (iv)}$$

2. If $\alpha_1 \left(= \sqrt{\frac{K_{a_1}}{c}}\right) \leq 0.1$, $x << c$

$\Rightarrow (c-x) \approx c$

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From eq (iii)

$$K_{a_1} = \frac{x^2}{c} \Rightarrow x = \sqrt{K_{a_1} \times c}$$

$$[H^+] = (x + y) = \sqrt{K_{a_1} \times c}$$

$$\boxed{pH = \frac{1}{2}\left(pK_{a_2} - \log c\right)} \rightarrow \text{Applicable when } \alpha \leq 0.1$$

Note :

for polyacidic Base,

$$\boxed{pOH = \frac{1}{2}\left(pK_{b_1} - \log c\right)} \rightarrow \text{when } \alpha \leq 0.1$$

Q. Find pH of 0.2 M aq H_2CO_3.

$K_{a_1} = 4 \times 10^{-6}$ } for H_2CO_3

$K_{a_2} = 2 \times 10^{-10}$ }

Soln.

$$pH = \frac{1}{2}\left(6 - \log 4 - \log 0.2\right)$$

$$= \frac{1}{2}\left(6 - 0.6 - [illegible] + 1 - 0.3\right)$$

$$= [illegible]$$

$$= \frac{1}{2}\left(7 - 0.9\right)$$

$$= 3.05 \text{ Ans.}$$

* Special Case of H_2SO_4 :

$$\underset{c}{H_2SO_4(aq)} \longrightarrow \underset{0}{H^+(aq)} + \underset{0}{HSO_4^-(aq)}, \quad K_{a_1} \to \text{very large}$$
$$(c+x) \qquad c-x$$

$$\underset{c-x}{HSO_4^-(aq)} \rightleftharpoons \underset{(x+c)}{H^+(aq)} + \underset{x}{SO_4^{2-}(aq)}, \quad K_{a_2} \approx 1.2 \times 10^{-2}$$

$$1.2 \times 10^{-2} = \frac{(x+c) \times x}{c-x} \qquad \text{—— (i)}$$

Solve Quadratic to find x.

Q $0.1\,M$, $H_2SO_4(aq)$, $K_{a_2} = 10^{-2}$
Find $[H^+]$ and pH

Soln $\underset{0}{H_2SO_4(aq)} \longrightarrow \underset{(c+x)}{H^+} + \underset{c-x}{HSO_4^-}$

$$\underset{c-x}{HSO_4^-} \rightleftharpoons \underset{(x+c)}{H^+} + \underset{x}{SO_4^{2-}}, \quad K_{a_2}$$

$$K_{a_2} = 0.01 = \frac{(x+0.1) \times x}{0.1-x}$$

$$x^2 + 0.11x - 0.001 = 0$$

$$x = \frac{-0.11 + \sqrt{(0.11)^2 + 4 \times 0.001}}{2}$$

$[H^+] = x + 0.1$, then find pH.

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* Salt Hydrolysis :-

Reaction of Salt with water.

$$\underset{\text{Salt}}{B^+A^-(aq)} + H_2O(l) \rightleftharpoons \underset{\text{acid}}{HA(aq)} + \underset{\text{Base}}{BOH(aq)}$$

(Reverse of Acid - Base neutralisation)

→ Salt hydrolysis is endothermic process, T↑, extent of hydrolysis of salt ↑

* Case - I

Salt of strong acid and strong Base
eg. NaCl, KBr, $BaCl_2$, $NaNO_3$, NaCl, etc.

eg. $NaCl(aq) + H_2O(l) \rightleftharpoons HCl(aq) + NaOH(aq)$

OR

$Na^+(aq) + Cl^-(aq) + H_2O(l) \rightleftharpoons H^+(aq) + Cl^-(aq) + Na^+(aq) + OH^-(aq)$

OR

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

$$K_a(H_2O) = 1.8 \times 10^{-16}$$

(extremely small)

→ No hydrolysis of NaCl (or Salt of S.A and S.B) and pH of such solution will be 7.

Imp. → No hydrolysis of cation of strong base and anion of strong acid.

* **Case - II**

Salt of WA and SB (case of anionic hydrolysis)

eg. $CH_3COO^-Na^+$, $HCOO^-K^+$, Na^+CN^- etc.

say, salt is NaA

HA	$A^-(aq)$	+ $H_2O(l)$	⇌ $HA(aq)$	+ $OH^-(aq)$
↓	c		0	0
A^-	$c(1-h)$		ch	ch

h → degree of hydrolysis.

K_h → Hydrolysis constant.

$$K_h = \frac{[HA][OH^-]}{[A^-]} \times \frac{[H^+]}{[H^+]}$$

$$\boxed{K_h = \frac{K_w}{K_{a(H.A)}}}$$ Remember.

$$K_h = \frac{ch \times ch}{c(1-h)} = \frac{ch^2}{(1-h)} \quad \text{— (i)}$$

Approximation :

1. If $h \left(= \sqrt{\frac{K_h}{c}}\right) \leq 0.1$, then $(1-h) \approx 1$

$K_h = ch^2$

$$h_{app.} = \sqrt{\frac{K_h}{c}}$$ → applicable when $h \leq 0.1$

Calculation of pH

$$[OH^-] = c \times h$$
$$= c \times \sqrt{\frac{K_h}{c}}$$
$$= \sqrt{\frac{K_w}{K_a}} \times c$$

$$[H^+] = \frac{K_w}{[OH^-]} = \sqrt{\frac{K_w \times K_a}{c}}$$

$$pH = \frac{1}{2}\left(pK_w + pK_a + \log c\right)$$ → applicable when $h \leq 0.1$

→ Such solutions are alkaline and turns red litmus blue.

* (Anionic Hydrolysis yields Basic Solution)

Note. → weaker is the acid greater will be h

$\because K_h = \frac{K_w}{K_A}$

$K_a \downarrow$ (i.e weaker acid), $h \uparrow$, $\therefore pH \uparrow$

Q. Which of the following solution (all 1M) will have highest pH.

A. Na_2S
B. Na_2O
C. NaCl
D. Na_2Se

$O^{2-} > S^{2-} > Se^{2-} > Te^{2-}$

decreasing order of extent of hydrolysis
∴ decreasing pH of solution

Ans. $H_2O < H_2S < H_2Se < H_2Te$

Increasing acidic strength

* Case III :

Salt of SA and WB : (case of cationic hydrolysis)

eg. NH_4Cl, NH_4Br, $C_6H_5NH_3Cl$ etc.

✓ ✗ ✓ ✗ ✓ ✗

{Final solution acidic turns B → R}

BCl ✓ ✗

$B^+(aq) + H_2O(l) \rightleftharpoons BOH + H^+$

c		0	0
c(1-h)		ch	ch

$$K_h = \frac{[BOH][H^+]}{[B^+]} \times \frac{[OH^-]}{[OH^-]}$$

$$K_h = \frac{K_w}{K_b(BOH)}$$

$$K_h = \frac{ch \times ch}{c(1-h)} = \frac{ch^2}{1-h} \quad \text{— (1)}$$

If $h \leq 0.1$, $(1-h) \approx 1$

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$\Rightarrow K_h = ch^2$

$$h = \sqrt{\frac{K_h}{c}} = \sqrt{\frac{K_w}{K_b \times c}} \rightarrow \text{when } h \leq 0.1$$

pH Calculation :

$$[H^+] = ch$$
$$= c \times \sqrt{\frac{K_w}{K_b \times c}} = \sqrt{\frac{K_w \times c}{K_b}}$$

$$pH = \frac{1}{2}\left(pK_w - pK_b - \log c\right) \rightarrow \text{when } h \leq 0.1$$

Q. Find pH, degree of hydrolysis, hydrolysis constant of 0.5 M aq. HCOONa solution $K_a(HCOOH) = 5 \times 10^{-5}$.

Soln $K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{5 \times 10^{-5}} = 0.2 \times 10^{-9}$

$= 2 \times 10^{-10}$.

$h = \sqrt{\frac{2 \times 10^{-10}}{0.5}} = 2 \times 10^{-5} <<< 0.1$

$pH = \frac{1}{2}\left(pK_w + pK_a + \log c\right)$

$= \frac{1}{2}\left(14 + (5 - \log 5) + \log 0.5\right)$

$= \frac{1}{2}(19 - 0.4 + 0.7 - 1)$

$= 18/2 = 9.$

Q. Find pH of 0.02 M aqueous solution of BCl salt if $K_b(BOH) = 10^{-12}$

Soln $B^+(aq) + H_2O(l) \rightleftharpoons BOH(aq) + H^+(aq)$

$K_h = \frac{10^{-14}}{10^{-12}} = 10^{-2}$

$h_{app.} = \sqrt{\frac{10^{-2}}{0.02}} = \sqrt{0.5} = 0.7$ ✗

$K_h = \frac{Ch^2}{1-h}$

$10^{-2} = \frac{2 \times 10^{-2} h^2}{(1-h)}$

$2h^2 + h - 1 = 0$

$h^2 + 0.5h - 0.5 = 0$

$h = \frac{-0.5 + \sqrt{0.25 + 4 \times 0.5}}{2}$

$h = 0.5$ ✓

$[H^+] = Ch$

$= 2 \times 10^{-2} \times 0.5 = 10^{-2}$

$\boxed{pH = 2}$ Ans.

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* **Case - IV :**

Salt of weak acid and weak Base :

eg. $CH_3\overset{-}{C}OO\overset{+}{N}H_4$, $\overset{+}{N}H_4\overset{-}{C}N$ etc.

$\overset{+}{B}\overset{-}{A}$ (s) $\longrightarrow$ c mol/lit

$\xrightarrow[H_2O(l)]{}$ $B^+(aq) + A^-(aq)$

$B^+(aq) + A^-(aq) + H_2O \rightleftharpoons BOH(aq) + HA(aq)$

$A^- + H_2O(l) \rightleftharpoons HA + OH^-$, $K_{h_1} = \frac{K_w}{K_a}$

$c(1-h_1)$

$B^+ + H_2O(l) \rightleftharpoons BOH + H^+$, $K_{h_2} = \frac{K_w}{K_b}$

$c(1-h_2)$

$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$

** $h_1 \approx h_2$ (approximation)

* When a compound is dissolved in water and both H^+ and OH^- are formed, then in such case extent of both the reactions forming H^+ and OH^- is taken nearly same.
Also pH of such solution does not depend on concentration of solution.
This kind of phenomenon is observed in

(i) Salt of WA & WB
(ii) Salt containing amphiprotic anion.
(iii) Amino acids (Zwetterion).

$$B^+(aq) + A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + BOH(aq)$$

	c	c		0	0
	$c(1-h)$	$c(1-h)$		ch	ch

$$K_h = \frac{[HA][BOH]}{[B^+][A^-]} \times \frac{[H^+][OH^-]}{[H^+][OH^-]}$$

$$\boxed{K_h = \frac{K_w}{K_a \times K_b}}$$

$$K_h = \frac{ch \times ch}{c(1-h) \times c(1-h)}$$

$$\boxed{K_h = \frac{h^2}{(1-h)^2}}$$

$$\Rightarrow \boxed{h = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}}$$

Calculation of pH :

$$K_h = \frac{[HA][BOH]}{[B^+][A^-]} = \frac{[HA]^2}{[A^-]^2}$$

$$HA \rightleftharpoons H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$\frac{[HA]}{[A^-]} = \frac{[H^+]}{K_a}$$

Now, use pt.

$$\frac{[H^+]^2}{K_a^2} = \frac{K_w}{K_a \times K_b}$$

$$[H^+] = \sqrt{\frac{K_w \times K_a}{K_b}}$$

$$\boxed{pH = \frac{1}{2}\left[pK_w + pK_a - pK_b \right]}$$ → pH of such solution is independent of concentration of solution.

- pH may be more than 7 (Basic), or less than 7 (acidic) or = 7 (neutral) depends on relative values of K_a and K_b.
- pH does not depend on concentration.

Q. Find pH of aqueous solution of 0.2 M NH_4CN. Given $K_{a(HCN)} = 10^{-8}$, $K_{b(NH_3)} = 10^{-5}$

Soln

$$pH = \frac{1}{2}(14 + 8 - 5)$$

$$= 8.5 \text{ (slightly alkaline)}$$

Q. How many of the following can turn blue litmus red ?

$BaCl_2$, $HCOONa$, NH_4Br, KCN, aniline, $Sr(NO_3)_2$, $HCOOH$, CH_3COONa, CH_3COONH_4

Soln. Acidic Solution :

NH_4Br, $HCOOH$

* Hydrolysis of Salts having polyvalent cation or polyvalent anion :

Polyvalent cation : Fe^{2+}, Co^{2+}, Al^{3+}, Fe^{3+} etc.

Polyvalent anion : CO_3^{2-}, S^{2-}, PO_4^{3-} etc.

(a) Salt having polyvalent anion :

eg. Na_2CO_3 (×, ✓), Na_2S (×, ✓), Na_2PO_4 (×, ✓) etc

Example

$$\underset{c-x}{A^{2-}(aq)} + H_2O(\ell) \rightleftharpoons \underset{x-y}{HA^-} + \underset{(x+y)}{OH^-}, \quad K_{h_1} = \frac{[HA^-][OH^-]}{[A^{2-}]}$$

$$\underset{x-y}{HA^-(aq)} + H_2O(\ell) \rightleftharpoons \underset{y}{H_2A} + \underset{(y+x)}{OH^-}, \quad K_{h_2} = \frac{K_w}{K_{a_2}}$$

$$\Rightarrow K_{h_1} = \frac{K_w}{K_{a_2}} = \frac{(x-y)(x+y)}{c-x} \quad \text{---(i)}$$

$$K_{h_2} = \frac{K_w}{K_{a_1}}$$

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$$K_{h_2} = \frac{K_w}{K_{a_1}} = \frac{y(x+y)}{x-y} \quad \text{---(ii)}$$

* Approximations :

(i) Since $K_{a_2} < K_{a_1} \Rightarrow K_{h_1} > K_{h_2}$

$\therefore y \ll\ll x$

$$K_{h_1} = \frac{K_w}{K_{a_2}} = \frac{x^2}{c-x}$$

$$K_{h_2} = \frac{K_w}{K_{a_1}} = y$$

(ii) If $h_1 \left(= \frac{\sqrt{K_{h_1}}}{c}\right) \leq 0.1$, $x \ll c$

$\Rightarrow (c-x) \simeq c$

Now, $K_{h_1} = \frac{K_w}{K_{a_2}} = \frac{x^2}{c}$

$$\Rightarrow x = \sqrt{\frac{K_w \times c}{K_{a_2}}}$$

$$[OH^-] = (x+y) = \sqrt{\frac{K_w \times c}{K_{a_2}}}$$

$$[H^+] = \frac{K_w \times K_{a_2}}{\sqrt{\frac{K_w \times c}{K_{a_2}}}}$$

$$[H^+] = \sqrt{\frac{K_w \times K_{a_2}}{c}}$$

$$\boxed{pH = \frac{1}{2}\left(pK_w + pK_{a_2} + \log c\right)} \rightarrow \text{when } h_1 \leq 0.1$$

for Na_3PO_4 or PO_4^{3-}

$$pH = \frac{1}{2}\left(pK_w + pK_{a_2} + \log c\right)$$ → when $h_1 \leq 0.1$

⇒ Similarly for bivalent cation (Fe^{2+}, Co^{2+} etc)

$$pH = \frac{1}{2}\left(pK_w - pK_{b_2} - \log c\right)$$ → when $h_1 \leq 0.1$

⇒ Trivalent cation (Fe^{3+}, Al^{3+})

$$pH = \frac{1}{2}\left(pK_w - pK_{b_3} - \log c\right)$$ → when $h_1 \leq 0.1$

Q. How many of the following aqueous solutions will turn red litmus blue?

Potash alum, $FeCl_3$, $Ba(NO_3)_2$, $C_6H_5NH_3Cl$, NH_4NO_3, CH_3COOK, $ZnCl_2$, Na_3PO_4, K_2CO_3

Soln. Basic Solution :-

CH_3COOK, Na_3PO_4, K_2CO_3

Eg. $NaHCO_3$ NaH_2PO_4 Na_2HPO_4

$pH = \frac{1}{2}\left(pK_1 + pK_2\right)$ $pH = \frac{1}{2}\left(pK_1 + pK_2\right)$ $pH = \frac{1}{2}\left(pK_2 + pK_3\right)$

* pH of salt containing amphiprotic anion :-

eg. $NaHCO_3$, NaH_2PO_4, Na_2HPO_4, $NaHS$ etc.

Example -

$$NaHCO_3(s) \xrightarrow{H_2O(l)} Na^+(aq) + HCO_3^-(aq)$$

$$HCO_3^- + H_2O(l) \rightleftharpoons CO_3^{2-} + H_3O^+ \quad \text{(acidic)}$$

$$HCO_3^- + H_2O(l) \rightleftharpoons H_2CO_3 + OH^- \quad \text{(basic)}$$

<u>pH of aq. $NaHCO_3$</u>

$$\underset{c-2x}{HCO_3^-}(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq), \quad K_{a_2} = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]} \quad -(i)$$

$$\underset{c-2x}{HCO_3^-} + H_2O(l) \rightleftharpoons H_2CO_3 + OH^-, \quad K_h = \frac{[H_2CO_3][OH^-]}{[HCO_3^-]} \times \frac{[H^+]}{[H^+]}$$

$$-(ii)$$

$$K_h = \frac{K_w}{K_{a_1}} \quad -(iii)$$

$$H^+ + OH^- \rightarrow H_2O(l)$$

→ In such case extent of both the reaction are assumed same.

$$[CO_3^{2-}] = [H_2CO_3]$$

(i) ÷ (ii), $\frac{K_{a_2} \times K_{a_1}}{K_w} = \frac{[H^+][CO_3^{2-}]}{[H_2CO_3][OH^-]}$

$$\Rightarrow [H^+] = \sqrt{K_{a_1} \times K_{a_2}} = \frac{[H^+]^2}{K_w}$$

$$\boxed{pH = \frac{1}{2}(pK_{a_1} + pK_{a_2})}$$

pH is independent of conc. of solution

Q. Which is (are) salt containing amphiprotic anion :

A. $NaHC_2O_4$
B. $NaHS$
C. NaH_2PO_2
D. Na_2HPO_3

Buffer Solution :-

The solution whose pH is only slightly changed when a small amount of an acid or base or water is added into it.

or

The solution which resist change in its pH on addition of a small amount of an acid or base or water.

eg. Animal Blood is also buffered due to dissolved CO_2 (i.e H_2CO_3) and $NaHCO_3$

eg. Aqueous solution of borax is also buffered.

$$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{H_2O(l)} B(OH)_3 + [B(OH)_4]^-$$

1 : 1 molar ratio.

* Types of Buffer Solution :
  - Mixed buffer
  - Simple buffer.

(a) Mixed buffer :

⊙ Acidic buffer :

A mixture of weak acid and its conjugate base.

eg. $CH_3COOH + CH_3COONa$

eg. $H_2CO_3 + NaHCO_3$

eg. $NaH_2PO_4 + Na_2HPO_4$

eg. $HCOOH + HCOONa$

⊙ Basic buffer :

A mixture of a weak base and its conjugate acid.

eg. $NH_4OH(aq) + NH_4Cl(aq)$

eg. $C_6H_5NH_2(aq) + C_6H_5NH_3^+ Br^-$

(b) Simple buffer :

Aqueous solution of a salt of weak acid and weak base.

eg. $NH_4Cl(aq)$, $CH_3COONH_4(aq)$

Note : → Strong acid and it's conjugate base do not act as Buffer Solution.

→ Strong Base and its conjugate acid do not form Buffer Solution.

* Buffer action :

eg. an acidic buffer (HA + NaA)

WA ↙ ↘ conjugate base

$CH_3COOH(aq) + CH_3COONa(aq)$

→ To this solution, a small amount of HCl is added.

$H^+ \longrightarrow$ Externally added.

$$CH_3COONa(aq) + H^+Cl^-(aq) \longrightarrow CH_3COOH(aq) + NaCl(aq)$$

$$K_{eq.} = \frac{[CH_3COOH]}{[CH_3COO^-][H^+]} = \frac{1}{K_a} = 10^{+5}$$

∵ K_{eq} is very large

∴ Reaction moving in forward direction.

→ To this solution, a small amount of base is added

$$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COO^-(aq) + H_2O(l) + Na^+$$

↑ externally added

$$K_{eq.} = \frac{[CH_3COO^-]}{[CH_3COOH][OH^-]} \times \frac{[H^+]}{[H^+]}$$

$$K_{eq.} = \frac{K_a}{K_w} \approx \frac{10^{-5}}{10^{-14}} \simeq 10^9 \text{ (very large)}$$

* pH of acidic buffer solution :

Eg. HA + NaA
a s mol/lit

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$, $K_a = \frac{x \times (s+x)}{a-x}$ — (i)
a 0 s
a-x x s+x

Approximation :

→ If K_a is small and a is high so that $\alpha_{HA} \leq 0.1$
$\alpha_{app} = \left(\frac{\sqrt{K_a}}{c}\right)$ $(x << a)$

$\Rightarrow (a-x) \approx a$

→ If concentration of acid and salt are enough high and comparable.

$(x << s) \Rightarrow (s-x) \approx s$

from eq (i)

$K_a = \frac{x(s+x)}{a-x} = [H^+] \times \frac{s}{a}$

$[H^+] = K_a \times \frac{a}{s}$

→ applicable when
(i) K_a is low.
(ii) a & s are high & comparable

$pH = pK_a + \log \frac{s}{a}$ or $pH = pK_a + \log \frac{[salt]}{[acid]}$

(Henderson - Hasselbalch Equation)

* p^H of acidic buffer may be less than 7, = 7, or more than 7.

* Similarly for Basic buffer,

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

OR

$$pOH = pK_b + \log \frac{s}{b}$$

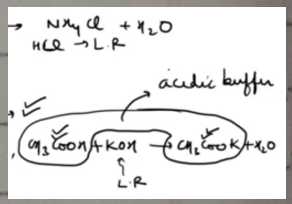
— applicable when
(i) K_b is low.
(ii) $\alpha \approx \infty$
conc. of base and salt are high & comparable.

(B.) Simple Buffer :
aq. solution CH_3COONH_4

$$pH = \frac{1}{2}\left(pK_w + pK_a - pK_b\right)$$

(IIT - Ad)
Q. Which of the following can act as buffer solution?

a. $NH_4OH + HCl$
b. $KNO_3 + HNO_3$
c. $Na_3PO_4 + Na_2HPO_4$
d. $CH_3COOH + KOH$



Q. Find mass of $(NH_4)_2SO_{4(s)}$ to be added into 0.1 M aq. NH_3 (500ml) to form a solution of $p^H = 9.3$. $[K_b(NH_3) = 2 \times 10^{-5}$

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Soln $\underbrace{NH_3 + NH_4^+}_{\text{Basic Buffer}}$ let $(NH_4)_2SO_4 \rightarrow w\,g$

$$n = \frac{w}{36+96} = \frac{w}{132} \text{ mol}$$

$$n_{NH_4^+} = \frac{w}{132} \times 2 = \frac{w}{66} \text{ mol}$$

$$pOH = pK_b + \log \frac{s}{b}$$

$$14 - 9.3 = 4.7 + \log \frac{(w/66) \text{ mol}}{0.1 \times 0.5 \text{ mol}}$$

$$1 = \frac{w}{66 \times 0.05}$$

$$\Rightarrow w = 66 \times \frac{1}{20} = 3.3\,g \text{ Ans.}$$

Q. 1 mol each of acetic acid and sodium acetate are dissolved in water to form 1 lit. solution, whose pH is found to be 4.7. Find pH if :

(i) 0.2 mol of HCl gas is passed into above buffer solution.

(ii) 0.2 mol of NaOH(s) is dissolved in the original buffer solution.

Soln. $pH = pK_a + \log \frac{1}{1}$ ...(i)

$4.7 = pK_a + \log \frac{1}{1}$

$\Rightarrow pK_a (CH_3COOH) = 4.7$

(i) $CH_3COONa + HCl(g) \longrightarrow CH_3COOH(aq) + NaCl(aq)$

1 (aq) 0.2 mol, L.R 1+0.2 = 1.2 mol

1 − 0.2 = 0.8

$pH = 4.7 + \log \frac{0.8}{1.2}$

$pH = 4.7 + \log \frac{2}{3} = 4.7 + 0.3 - 0.48$

$pH = 4.52$

(ii) $CH_3COOH(aq) + NaOH(s) \longrightarrow CH_3COONa + H_2O$

1 0.2 1

1 − 0.2 = 0.8 1.2 mol

$pH = 4.7 + \log \frac{1.2}{0.8}$

$pH = 4.7 + 0.48 - 0.3$

$pH = 4.88$ Ans.

(*) Buffer Capacity :-
(only for IIT JEE-Advanced)

→ This term is used to compare effectiveness of buffer solutions.

- It is the number of moles of an acid or base to be added into 1 lit of buffer solution to change it's pH by 1 unit, keeping $\underbrace{(a+s)}_{\text{acidic Buffer}}$ or $\underbrace{(b+s)}_{\text{Basic Buffer.}}$ constant.

For acidic buffer, buffer capacity $= \frac{-da}{d(pH)} = \frac{ds}{d(pH)}$

for basic buffer, buffer capacity $= \frac{db}{d(pH)} = \frac{-ds}{d(pH)}$

eg. acidic buffer,

$$\left.\begin{matrix} HA & + & NaA \\ a & & s \end{matrix}\right\} \quad a+s = K \text{ (constant)}$$

$$pH = pK_a + \log \frac{s}{a}$$

$$pH = pK_a + \log \frac{s}{(K-s)}$$

$$\frac{d(pH)}{ds} = 0 + \frac{1}{2.303} \times \frac{1}{\left(\frac{s}{K-s}\right)} \left[\frac{(K-s)\times 1 - s(-1)}{(K-s)^2}\right]$$

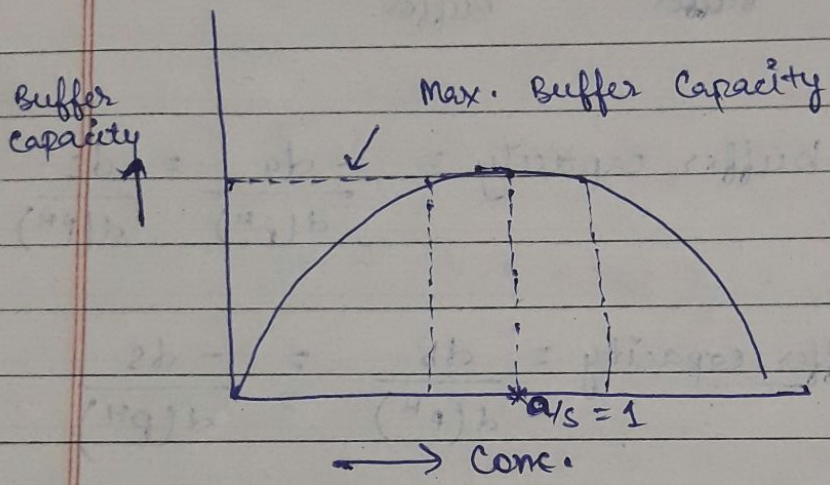
$$\left\{\frac{ds}{d(pH)} = \frac{2.303\, s(K-s)}{s(K-s)} \text{ or } = 2.303\,(s)\right\}$$

* For maxima of buffer capacity,

$$\frac{d}{ds}(\text{Buffer capacity}) = 0$$

$$\frac{2.303}{K}\left[K - 2s\right] = 0 \Rightarrow s = K/2$$

$$\Rightarrow \boxed{a = s = K/2}$$ for max. buffer capacity.



→ for most effective → Buffer (acidic), $a = s$

→ Basic Buffer, $b = s$

→ For an effective buffer,

$$0.1 \leq \frac{s}{a} \leq 10 \quad \text{(for acidic buffer)}$$

$$0.1 \leq \frac{s}{b} \leq 10 \quad \text{(for basic buffer)}$$

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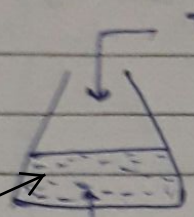
* Acid - Base Titration :-



$HCl_{(aq)} + NaOH_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(l)}$

→ Phenolphthalein (HPh), colourless ← 8 to 10 → more pink

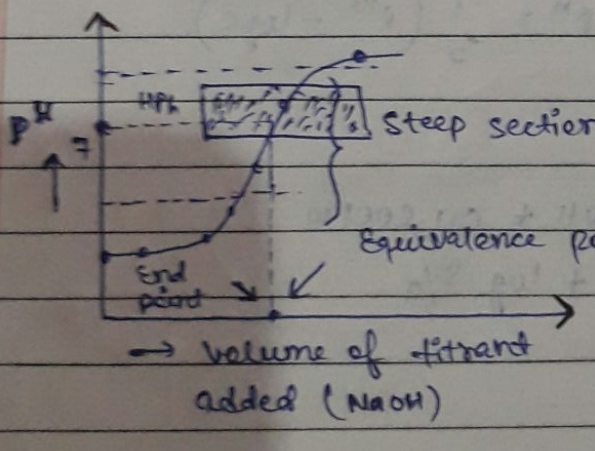
pH transition range.

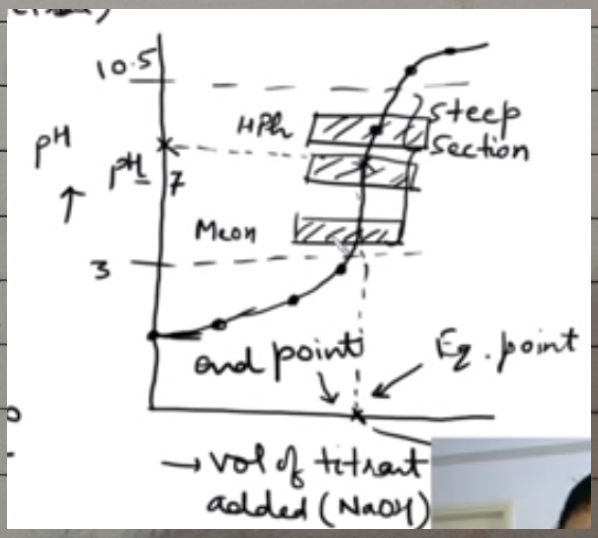
→ Methyl orange (MeOH) Red ← 3.1 to 4.4 → yellow

* Case I :

Titration of SA vs SB :

↑ HCl (aq) ↑ NaOH (titrant)





At start : $[H^+] = [HCl]$ ✓

$\therefore pH =$ ✓

$[HCl]_0 = 0.1\,M, 100\,ml$

$[NaOH] = 0.1\,M, V$

Before eq. point :

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

$0.1 M \times 100 = 10$ mmol ; $0.1 \times V$ mmol (L.R)

$$[H^+]_f = \frac{(10 - 0.1 \times V)}{(100 + V)}, \text{ here } V < 100\,ml$$

At equivalence point :-

$$\underset{\times}{HCl} + \underset{\times}{NaOH} \longrightarrow \underline{NaCl} + H_2O$$

---

* Case - II

Titration of WA Vs SB :

$$CH_3COOH(aq) + \underset{\text{titrant}}{NaOH(aq)} \longrightarrow CH_3COONa(aq) + H_2O(l)$$

→ At start : CH_3COOH ✓, $pH = \frac{1}{2}(pK_a - \log c)$

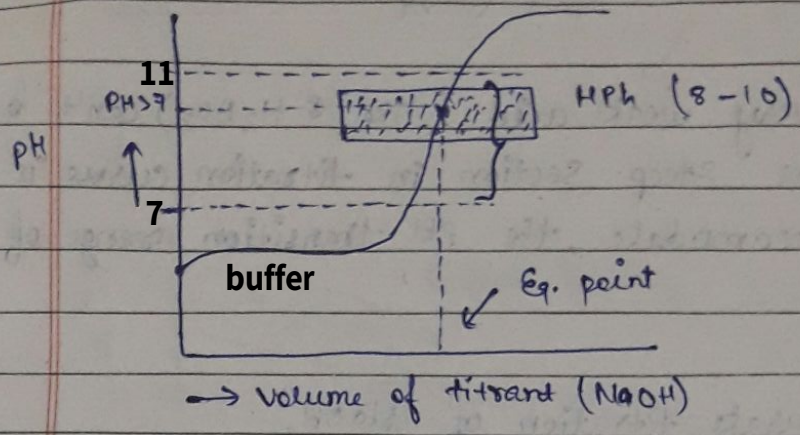
→ Before eq. point, $CH_3COOH + CH_3COONa$

$$pH = pK_a + \log \frac{S}{a}$$

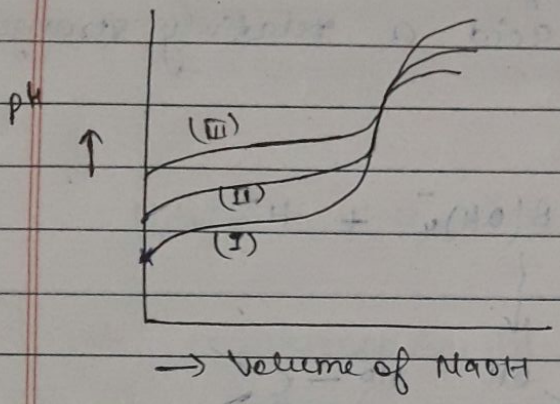
→ At eq. point

$CH_3COONa(aq)$

$$pH = \frac{1}{2}(pK_w + pK_a + \log c)$$



Imp. Q



HA, HB, HC

All have same initial concentration

I → HC

II → HB

III → HA

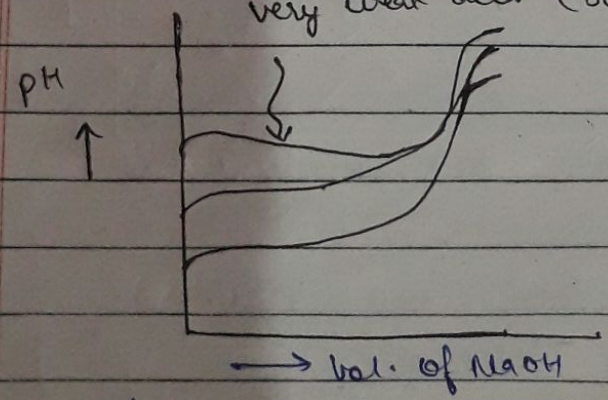
Compare Ka values of HA, HB, HC.

Soln. HC > HB > HA

decreasing Ka value.

III X (X)

NOTE :

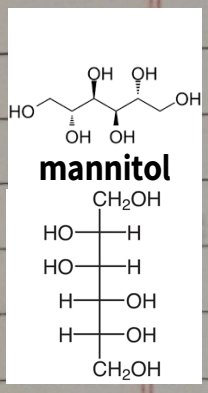


(Titration of curve of WA Vs BSB)

$B(OH)_3 + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+$

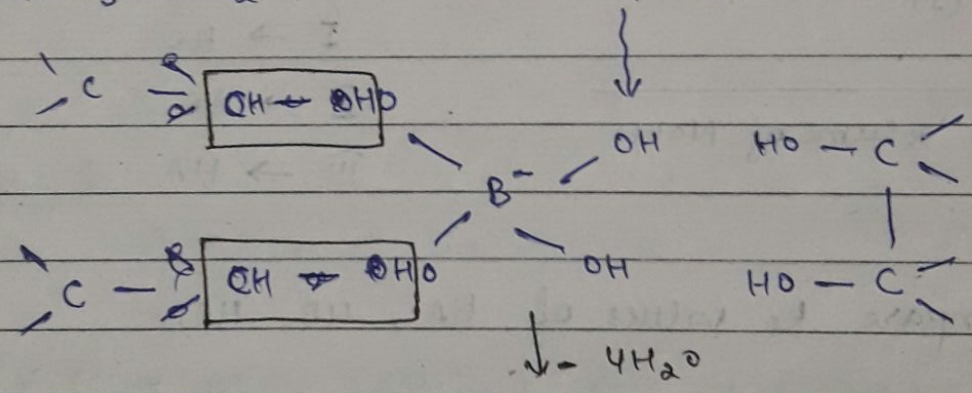
* Titration of a very weak acid (like H_3BO_3) can't be done with NaOH as steep section in titration curve is too small to accomodate the PH transition range of indicator.

→ To do the accurate titration of NaOH, cis-1,2 diol (eg. ethylene, glycol, glycerol, mannitol, cathecol, salycyclic acid, etc) are added which make boric acid a relatively stronger acid.

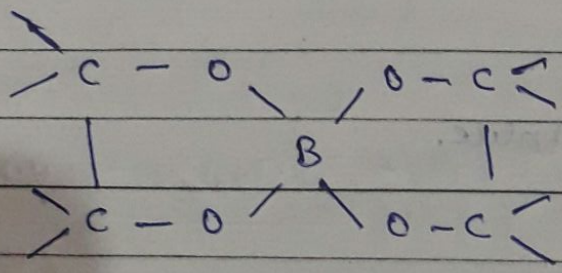


⇒

$B(OH)_3 + H_2O(l) \rightleftharpoons B(OH)_4^- + H^+$



$\downarrow -4H_2O$

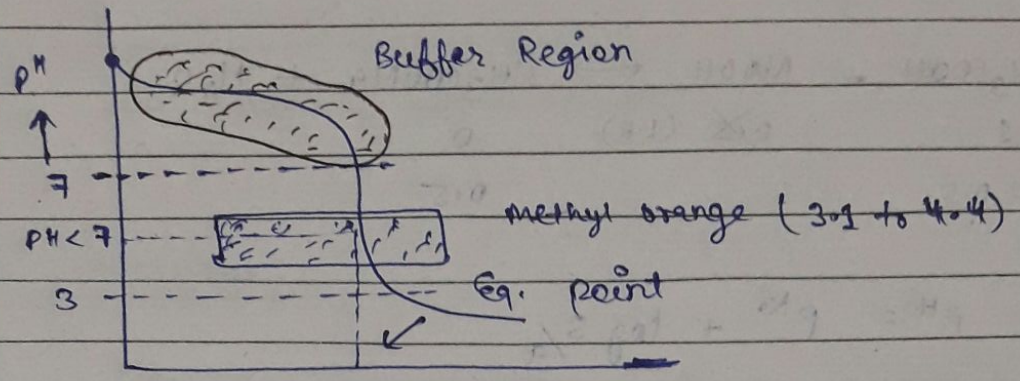


Complex

by Le Chatelier principle, the acidic strength of boric acid is increased in presence of cis-1,2-diols and hence, it can be titrated with NaOH

Case III Titration of Weak Base V/s Strong Acid

NH_4OH — HCl (aq) (titrant)



$$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$$

At equivalence point, Solution contain NH_4Cl (aq) i.e cationic hydrolysis

$$pH = \frac{1}{2}(pK_w - pK_b - \log c) < 7$$

→ Half eq. point

eg. $NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$

a ✓ 0

a/2 a/2

→ 1/5th eq. point

$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$

a ✓ 0

$\frac{4a}{5}$ $\frac{1}{5}a$

Q In 1 lit. 1M CH_3COOH + 1 lit, 0.5 M NaOH is added
Find pH
[pK_a (CH_3COOH) = 4.7]

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

1 — 0.5 (L.R) — 0

1 − 0.5 = 0.5 — 0.5

$pH = pK_a + \log S/a$

$= 4.7 + \log \frac{0.5}{0.5} = 4.7$

(3) Case – IV

Titration of WA v/s WB :

- At start, CH_3COOH } CH_3COOH — NH_4OH (aq) (titrant)

BTEP

- $CH_3COOH + NH_4OH \longrightarrow CH_3COO^-NH_4^+ + H_2O$
(L.R)

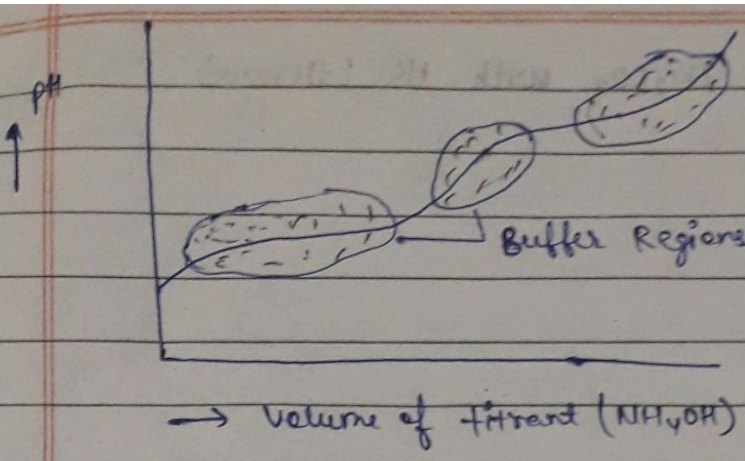
$CH_3COOH + CH_3COO^- + NH_4^+$

Buffer

- At eq. point

$CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O$
X X ✓

Salt contains only $CH_3COO^-NH_4^+$ (Simple Buffer)



∴ No suitable indicator.

Conclusion:

Titration ~~curve~~ using indicator cannot be satisfactorily done, in case of weak acid and weak Base.

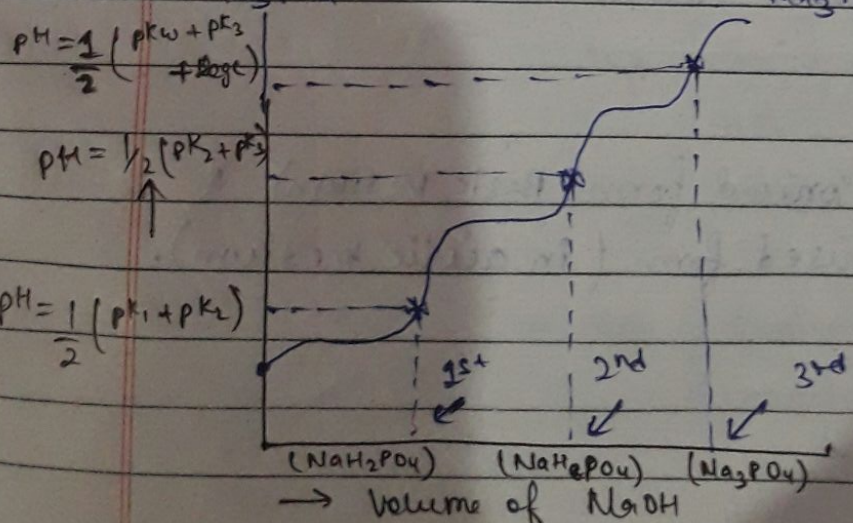
→ Such titration is very accurately done by conductometric titration.

* Titration of H_3PO_4 v/s $NaOH$ (titrant)

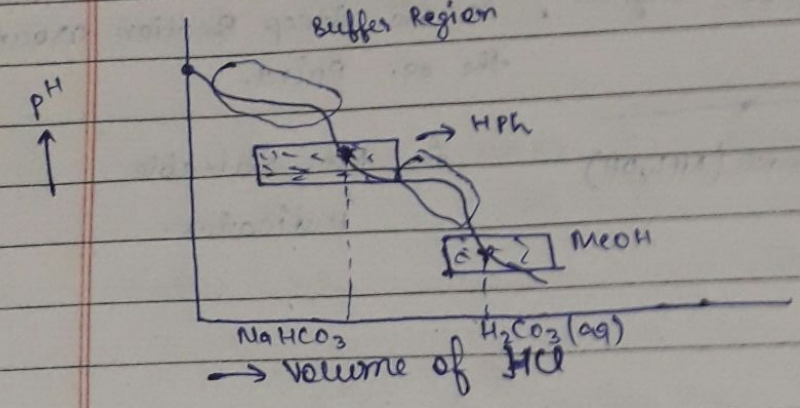
$H_3PO_4 + NaOH \longrightarrow NaH_2PO_4 + H_2O$ (1^{st} eq. point)

$H_3PO_4 + 2NaOH \longrightarrow Na_2HPO_4 + 2H_2O$ (2^{nd} eq. point)

$H_3PO_4 + 3NaOH \longrightarrow Na_3PO_4 + 3H_2O$ (3^{rd} eq. point)



* Double titration of Na_2CO_3 with HCl (strong)



(*) Theory of Indicators :-

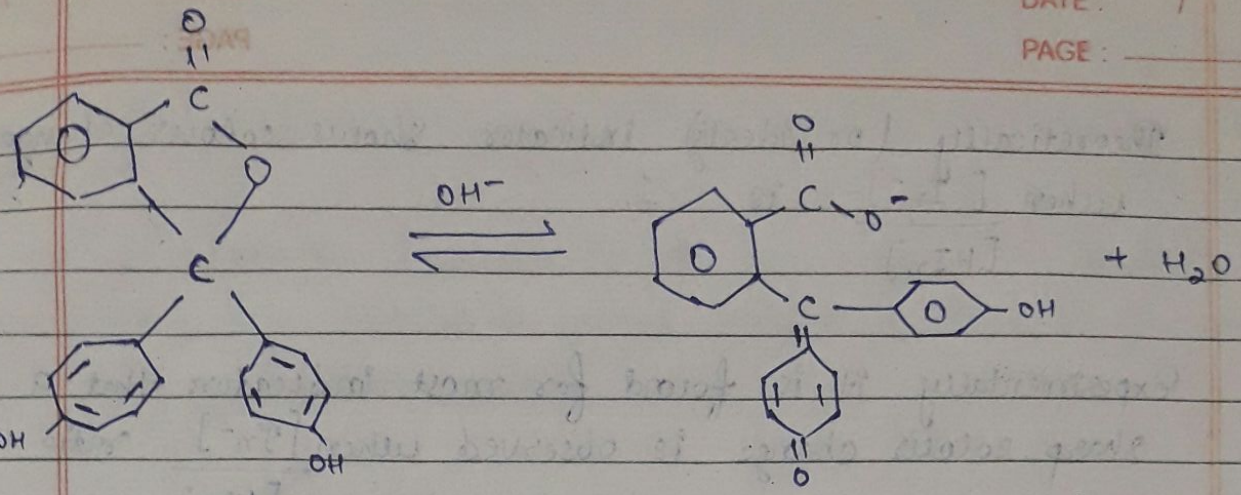
→ To detect the equivalence point, indicators are used.

→ for acid - base titrations, indicators used are weak organic acids (Phenolphthalein (HPh)) and a weak organic base (Methyl orange, (MeOH)).

→ During the process of titration, the extent of ionisation of indicator do change as pH of solution changes.

→ Indicator shows different colours due to their ionised and unionised form.

eg. HPh is pink in ionised form (Basic medium) & colourless in unionised form (in acidic medium).



Colourless in unionised form (Benzenoid form)

Pink in ionised form. (quinonoid form)

→ Methyl orange (a weak base indicator) (4.4 to 3.3)

Red. ← 3.3 to 4.4 → yellow

eg. a weak organic acid indicator (HIn)

$$HIn(aq) \rightleftharpoons H^+(aq) + In^-(aq), \quad K_{In} = \frac{[H^+][In^-]}{[HIn]}$$

$$[H^+] = K_{In} \times \frac{[HIn]}{[In^-]}$$

$$\boxed{pH = pK_{In} + \log \frac{[In^-]}{[HIn]}} \quad \text{——— (1)}$$

OR

$$\left\{ pH = pK_{In} + \log \frac{[\text{Ionised}]}{[\text{Unionised}]} \right\}$$

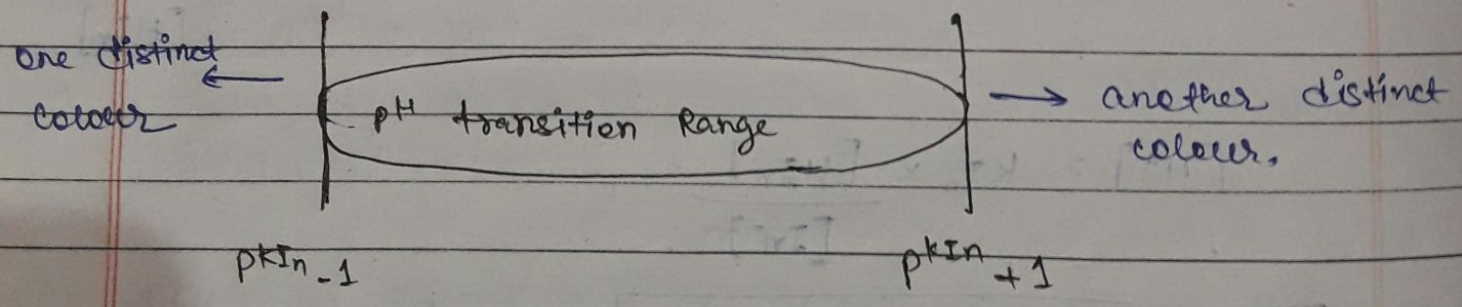
→ Theoretically (or ideally) indicator shows colour change when $\frac{[In^-]}{[HIn]}$ is 1.

→ Experimentally it is found for most indicators that a sharp colour change is observed when $\frac{[In^-]}{[HIn]}$ ratio turns from 0.1 to 10.

→ We see a distinct colour when $\frac{[In^-]}{[HIn]}$ ratio is 0.1 or less and another colour when this ratio is 10 or more.

→ When $[In^-]/[HIN]$ is 0.1, then $pH = pK_{In} + \log 0.1$
$= pK_{In} - 1$

When $[In^-]/[HIN]$ is 10, then $pH = pK_{In} + \log 10$
$= pK_{In} + 1.$



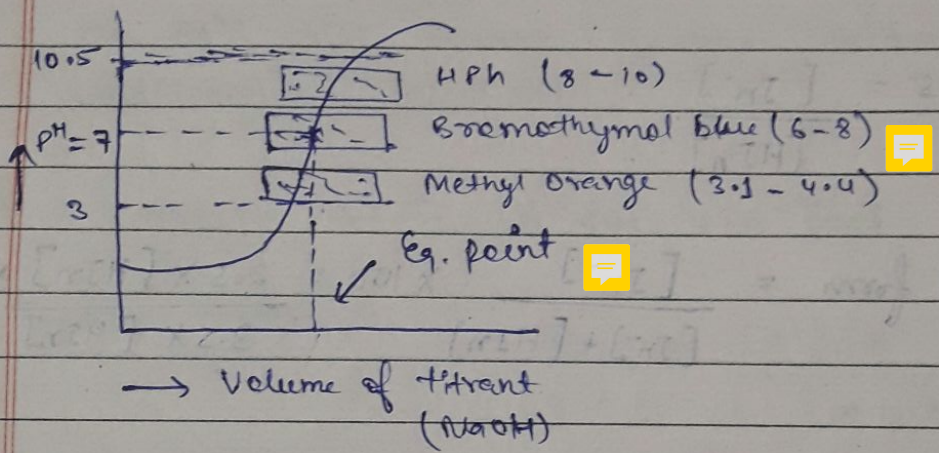
(Hph, $pK_{In} = 9$)

* Selection of Indicator :

"A suitable indicator is the one whose pH transition range lies within the steep section of titration curve near the equivalence point."

eg. titration of SA vs SB

↙ HCl (aq) ↘ NaOH (titrant)



→ In above SA vs SB titration, any of the three indicators shown may be used. However the best indicator here is bromothyl blue.

→ Best indicator is the one whose p^{KIn} is as close as possible to the pH at equivalence point.

Q pH at eq. point in an acid-base titration is 5. Select the best indicator ?

A. $pK_{In} = 3$

B. $pK_{In} = 9$

C ✓ $pK_{In} = 4.8$

D. $pK_{In} = 7$

Q An acid indicator has $pH = 2$, when its ionised form is 50%. Find its % ionised form when $[H^+] = 4 \times 10^{-3}$ M

Sol^n $$pH = 2 = pK_{In} + \log \frac{1}{1}$$

$$\Rightarrow pK_{In} = 2$$

$$3 - \log 4 = 2 + \log \frac{[In^-]}{[HIn]}$$

$$2.5 = \frac{[In^-]}{[HIn]}$$

$$\% \text{ ionised form} = \frac{[In^-]}{[In^-] + [HIn]} \times 100 = \frac{2.5 \times [HIn]}{3.5 \times [HIn]} \times 100$$

$$= \frac{500}{7} \%$$

* Solubility and Solubility Product :-

Solubility (S) :

It is the molar concentration of a substance in a solvent at Saturation Equilibrium at a given temperature.

{unit : Mol / lit}

* Soluble Salt → having solubility in water > 0.1 M.
Slightly Soluble Salt → $0.01 M \leq S < 0.1 M$
Sparingly Soluble salt → when $S < 10^{-2}$ M.
(or almost insoluble salt)

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eg. Soluble Salt : NaCl, KCl, K_2CO_3, Na_2CO_3, $AgNO_3$, KI etc.

eg. Sparingly Soluble Salt : AgCl, AgBr, AgI, Ag_2CrO_4, PbI_2, $BaSO_4$, $BaCrO_4$, $CaCO_3$, $SrCO_3$ etc.

eg. Slightly Soluble Salt :
BaS, CH_3COOAg, $AgNO_2$ etc.

→ When a sparingly soluble salt is dissolved in water, an eqm exist between formed aqueous ions and undissociated solid salt.

1:1 salt

$AB(s) \rightleftharpoons A^+(aq) + B^-(aq)$.

↑ Strong electrolyte

$$K_{eq} = \frac{[A^+][B^-]}{[AB_{(s)}]}$$

$$K_{sp} = [A^+][B^-]$$

↑ Solubility product of Salt.
(depends on temperature, independent of concentration)

$AB_2(s) \rightleftharpoons A^{2+}(aq) + 2B^-(aq)$

1:2 type salt

$$K_{sp(AB_2)} = [A^{2+}]_{eq} \times [B^-]^2_{eq}$$

* Solubility in Pure Waters :

eg. $AB(s) \rightleftharpoons A^+(aq) + B^-(aq)$

S mol/lit — S mol/lit — S mol/lit

$K_{sp} = [A^+][B^-]$

$= S \times S$

$$S = \sqrt{K_{sp}} \text{ mol/lit}$$

Note :
Every sparingly soluble salt
↓
Strong Electrolyte

eg. $PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq)$

S — S — 2S

$K_{sp} = [Pb^{2+}] \times [I^-]^2$

$= S \times (2S)^2$

$= 4S^3$

$$S = \left(\frac{K_{sp}}{4}\right)^{1/3}$$

A general Case :

$A_xB_y(s) \rightleftharpoons xA^{y+}(aq) + yB^{x-}(aq)$

S — xS — yS

$K_{sp} = [A^{y+}]^x \times [B^{x-}]^y$

$= (xS)^x \times (yS)^y$

$\Rightarrow K_{sp} = x^x y^y S^{x+y}$

$\Rightarrow$

$$S = \left(\frac{K_{sp}}{x^x y^y}\right)^{1/x+y}$$

* Solubility in presence of common ion :

→ In presence of common ion, solubility of a sparingly soluble salt is decreased.

eg. $PbI_2(s)$ is dissolved in aq. KI solution (c mol/lit)

$$\underset{s}{PbI_2(s)} \rightleftharpoons \underset{s}{Pb^{2+}(aq)} + \underset{(2s+c)}{2I^-(aq)}$$

$2s \ll\ll c$

$$K_{sp} = [Pb^{2+}][I^-]^2$$

$$\Rightarrow K_{sp} = s \times (c)^2$$

from here, calculate 's'.

Q. Find Solubility of salt Ag_2CrO_4 in pure water
$(K_{sp}(Ag_2CrO_4)) = 4 \times 10^{-12}$

Sol^n.

$$\underset{s}{Ag_2CrO_4(s)} \rightleftharpoons \underset{2s}{2Ag^+(aq)} + \underset{s}{CrO_4^{2-}(aq)}$$

$$(2s)^2 \times s = K_{sp}$$
$$4s^3 = 4 \times 10^{-12}$$
$$s = 10^{-4} M \text{ Ans.}$$

Q. Find Solubility of PbI_2 in 0.1M aq. solution of $Pb(NO_3)_2$ (aq)
$[K_{sp}(PbI_2)] = 10^{-15}$.

Sol^n.:

$$\underset{s}{PbI_2(s)} \rightleftharpoons \underset{(s+0.1)}{Pb^{2+}(aq)} + \underset{2s}{2I^-(aq)}$$

$0.1 \times (2s)^2 = 10^{-14}$

$4s^2 = 10^{-14}$

$s = \sqrt{\frac{1}{4} \times 10^{-14}}$

$s = \frac{1}{2} \times 10^{-7}\ M = 5 \times 10^{-8}\ M$ Ans.

Q. Calculate solubility of $Fe(OH)_3$ in pure water.
$[K_{sp}\ (Fe(OH)_3 = 1 \times 10^{-38}]$

Soln:

$Fe(OH)_3(s) \rightleftharpoons Fe^{3+}(aq) + 3OH^-(aq)$

$s \qquad\qquad s \qquad\qquad (3s + 10^{-7}\ M)$

$s \times (10^{-7})^3 = 10^{-38}$

$s = \frac{10^{-38}}{10^{-21}}$

$\boxed{s = 10^{-17}\ M}$

(*) Simultaneous Solubility:

→ When two sparingly soluble salts are present in same solution furnishing a common ion.

eg. $AgX(s) \rightleftharpoons Ag^+(aq) + X^-(aq)$, $K_{sp_1} = s_1(s_1+s_2)$

$s_1 \qquad\quad s_1+s_2 \qquad\quad s_1$

$AgY(s) \rightleftharpoons Ag^+(aq) + Y^-(aq)$, $K_{sp_2} = s_2(s_1+s_2)$

$s_2 \qquad\quad s_2+s_1 \qquad\quad s_2$

Q. Find solubility (simultaneous) of $BaSO_4$ and $BaCrO_4$ present in water in excess amount.

K_{sp} ratio of $BaSO_4$ and $BaCrO_4$ is 1 : 2 and $[Ba^{2+}]$ conc. at saturation eqm is 2×10^{-5} M.

Sol^n:

$$\underset{S_1}{BaSO_4(aq)} \rightleftharpoons \underset{S_1+S_2}{Ba^{2+}(aq)} + \underset{S_1}{SO_4^{2-}(aq)}$$

$$\underset{S_2}{BaCrO_4(aq)} \rightleftharpoons \underset{S_2+S_1}{Ba^{2+}(aq)} + \underset{S_2}{CrO_4^{2-}(aq)}$$

$$\frac{K_{sp_1}}{K_{sp_2}} = \frac{(S_1+S_2) \times S_1}{(S_2+S_1) \times S_2}$$

$$\frac{1}{2} = \frac{S_1}{S_2} \Rightarrow S_2 = 2S_1$$

$$[Ba^{2+}] = S_1 + S_2 = 2 \times 10^{-5} \text{ M}$$

$$= 3S_1 = 2 \times 10^{-5} \text{ M}$$

Simultaneous solubility.

$$S_1 = \frac{2}{3} \times 10^{-5} \text{ M}$$

$$S_2 = \frac{4}{3} \times 10^{-5} \text{ M}$$

Ans.

* ~~Applic~~

(*) Applications of K_{sp} :

1. Simple Precipitation :

$$AB(s) \rightleftharpoons A^+(aq) + B^-(aq)$$

At any instant a term called Solubility quotient or ionic product (Q_{ip}) can be defined as,

$$Q_{ip} = [A^+]_t \times [B^-]_t$$

* If $Q_{ip} < K_{sp}$, solution is unstaturated and more salt will dissolve to reach to saturation point.

* If $Q_{ip} > K_{sp}$, solution is supersaturated ~~and~~ (unstable) and precipitation will occur till,

$$Q_{ip} = K_{sp}.$$

* If $Q_{ip} = K_{sp}$, then saturation eqm. No more salt will dissolve.

2. Equal volumes of 1.5×10^{-5} M $AgNO_3(aq)$ and 1×10^{-5} M $KCl(aq)$ are mixed. Predict whether precipitation will occur or not.

$$K_{sp}(AgCl) = 10^{-10}$$

Soln:

$$Q_{ip} = [Ag^+]_t \times [Cl^-]_t$$

$$= \left(\frac{1.5 \times 10^{-5}\ M}{2}\right) \times \left(\frac{10^{-5}\ M}{2}\right)$$

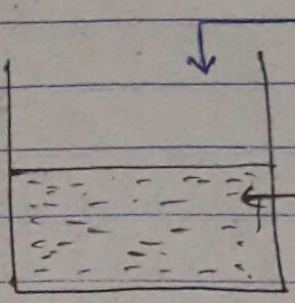
$$= 0.75 \times 0.5 \times 10^{-10} < K_{sp}$$

No, precipitation will occur.

13/11/12

2. Selective Precipitation :

Q.



$\left[K_{sp}(AgCl) = 10^{-10}\right]$, $K_{sp}(AgBr) = 10^{-14}$, $(AgI) = 10^{-18}$

(i) find the order of start of precipitation of I^-, Br^-, Cl^-

(ii) find conc. of I^- ion in solution when Br^- starts precipitati

(iii) find % of I^- precipitated as AgI when Br^- starts precipitation.

Sol^n :→ ppt. formation starts when Qip exceeds Ksp (i.e $Q_{ip} > K_{sp}$

→ for Cl^-, $[Ag^+]_{for\,Cl^-} \times 0.01 \geq 10^{-10}$

$\Rightarrow [Ag^+]_{for\,Cl^-} \geq 10^{-9}\,M$

→ for Br^- $[Ag^+]_{for\,Br^-} \times 0.1 \geq 10^{-14}$

$\Rightarrow [Ag^+]_{for\,Br^-} \geq 10^{-13}\,M.$

→ for I^-, $[Ag^+]_{for\,I^-} \times 0.1 \geq 10^{-18}$

$\Rightarrow [Ag^+]_{for\,I^-} \geq 10^{-17}\,M.$

(i) $I^- > Br^- > Cl^-$ (→)

order of start of precipitation in solution.

(ii) when Br^- starts precipitation,

$[Ag^+] = 10^{-13}$ M.

$10^{-13} \times [I^-] = 10^{-18}$

$[I^-] = 10^{-5}$ M.

(iii) % of I^- precipitated when Br^- starts precipitation

$$= \frac{(0.1 - 10^{-5})}{0.1} \times 100$$

$$\approx 99.9..\ \%\ (\text{approx})$$

(*) Solubility of sparingly soluble salt in suitable buffer solution :-

Q. Solubility of ~~pure~~ $Pb(OH)_2$ in pure water is $\frac{1}{3} * 10^{-5}$ M.

Find solubility of $Pb(OH)_2$ in a buffer solution of pH = 8.

Soln :

$$Pb(OH)_{2(s)} \rightleftharpoons Pb^{2+}_{(aq)} + 2OH^-_{(aq)}$$

$$s \qquad\qquad s \qquad\quad 2s$$

$$K_{sp} = 4s^3 = 4 \times \left(\frac{1}{3} \times 10^{-5}\right)^3$$

$K_{sp} = \frac{4}{27} \times 10^{-15}$

$Pb(OH)_2(s) \rightleftharpoons Pb^{2+} + 2OH^-$

$s' \qquad\qquad s' \qquad 2s' \quad 10^{-6}\,M.$

$\frac{4}{27} \times 10^{-15} = s' \times (10^{-6})^2$

$\Rightarrow s' = \frac{4}{27} \times 10^{-3}\,M$ Ans.

(*) Complexation Equilibrium :

→ Many sparingly soluble salts like AgCl, AgBr, AgI etc become soluble in presence of complexing ligands like NH_3, CN^-, $S_2O_3^{2-}$ etc.

eg. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, K_{sp}

$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$,

K_f = very large

↓

Formation or ~~Solu~~ Stability constant of complex.

on adding,

$AgCl(s) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$

$K_{eq} = K_{sp} \times K_f$

→ Solubility of a sparingly soluble salt increase highly in presence of complexing ligand.

→ $Ag^+(aq) + NH_3(aq) \rightleftharpoons [Ag(NH_3)]^+$, K_{f_1} (1st formation constant)

$[Ag(NH_3)]^+ + NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+$, K_{f_2} (2nd formation constant)

on, adding.

$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+$, $K_f = K_{f_1} \times K_{f_2}$ (formation constant.)

$$K_f = \frac{[Ag(NH_3)_2]^+}{[Ag^+][NH_3]^2}$$

→ $[Ag(NH_3)_2]^+(aq) \rightleftharpoons Ag^+(aq) + 2NH_3(aq)$,

K diss. (or) K instability.

$$= \frac{[Ag^+][NH_3]^2}{[Ag(NH_3)_2]^+} = \frac{1}{K_f}$$

eg. AgBr, AgI, AgCl are soluble in aq. KCN or hypo solution ($Na_2S_2O_3$)

→ $AgBr\downarrow + 2KCN(aq) \rightleftharpoons [Ag(CN)_2]^-(aq) + 2K^+(aq) + Br^-(aq)$ (CN = 2)

→ $AgI\downarrow + 2Na_2S_2O_3(aq) \rightleftharpoons [Ag(S_2O_3)_2]^{3-}(aq) + 4Na^+ + I^-$ (CN = 2)

Imp.

Q. Equal volumes of 0.2 M KCN and 0.06 M $AgNO_3$ are mixed. Find conc. of $[Ag^+]$ in final solution if

$[Ag(CN)_2]^-_{(aq)} \rightleftharpoons Ag^+(aq) + 2CN^-(aq)$,

$K_{diss.} = 4 \times 10^{-19}$

Sol^n

$[CN^-] = 0.1\ M$

$[Ag^+] = 0.03\ M$

Ag^+	+ $2CN^-$	$\rightleftharpoons$	$[Ag(CN)_2]^-$
0.03M	0.1M		0
	(0.1 − 0.06) = 0.04M		0.03M

$K_f = \frac{1}{4 \times 10^{-19}}$ (Very large)

$K_f = \frac{1}{4} \times 10^{19} = \frac{3 \times 10^{-2}}{[Ag^+] \times (4 \times 10^{-2})^2}$

$[Ag^+] = \frac{3 \times 10^{-19} \times 10^2}{4} = 0.75 \times 10^{-17}$

$[Ag^+] = 7.5 \times 10^{-18}\ M$ Ans.

IIT-AS.

Q. Find solubility of AgCl in aq. NH_3 (0.2M)

$K_{sp}(AgCl) = 10^{-10}$

$K_f\ [Ag(NH_3)_2]^+ = 10^9$

Sol^n

$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, $K_{sp} =$

S S S

$Ag^+ + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+(aq)$.

0.2 (0.2 − 2s) 0 s mol/lit

$$K_f = 10^9 = \frac{S}{[Ag^+] \times (0.2-2S)^2}$$

$$10^9 = \frac{S}{\frac{K_{sp}}{S} \times (0.2-2S)^2}$$

$$\Rightarrow \quad S = 3.9 \times 10^{-2}\ M.$$

→ AgCl solubility in pure water,

$$S' = \sqrt{K_{sp}} = 10^{-5}\ M.$$

Hence, solubility is increased

(*) Effect of Hydrolysis on solubility of Sparingly Soluble Salt.

$$MA(s) \rightleftharpoons M^+(aq) + A^-(aq)$$

S S S $\quad K_{sp} = S \times [A^-]$ — (1)

A^- is the anion of weak acid HA.

$$A^-(aq) + H_2O \rightleftharpoons HA(aq) + OH^-(aq)$$

$$K_h = \frac{K_w}{K_a} = \frac{[HA][OH^-]}{[A^-]}$$

$S = [A^-]_t + [HA]_t$ (Mass balance or Mol balance of A)

$$HA \rightleftharpoons H^+ + A^- \quad K_a = \frac{[H^+][A^-]}{[HA]}$$

$$[HA] = \frac{[H^+][A^-]}{K_a}$$

$$S = [A^-] + \frac{[H^+] \times [A^-]}{K_a(HA)}$$

$$\Rightarrow [A^-] = \frac{S}{1 + \frac{[H^+]}{K_a}}$$

From eq (1)

$$K_{sp} = S \times [A^-]$$

$$= S \times \frac{S}{1 + \frac{[H^+]}{K_a}}$$

$$\boxed{S = \sqrt{K_{sp}\left(1 + \frac{[H^+]}{K_a}\right)}}$$

→ At lower pH { solubility is increased.

Equal volumes of 0.02 M $AgNO_3(aq)$ and 0.02 M $HCN(aq)$ are mixed, find $[Ag^+]$ at eqm. $K_a(HCN) = 10^{-10}$, $K_{sp}(AgCN) = 10^{-16}$

Sol^n: $[Ag^+] = 0.01M$, $[HCN] = 0.01M$

$$Ag^+(aq) + HCN(aq) \rightleftharpoons AgCN\downarrow + H^+(aq)$$

$$K_{eq} = \frac{[H^+]}{[Ag^+][HCN]} \times \frac{[CN^-]}{[CN^-]} = \frac{K_a(HCN)}{K_{sp}(AgCN)}$$

$$= \frac{10^{-10}}{10^{-16}} = 10^6$$

$\Rightarrow$

$Ag^+ + HCN \rightleftharpoons AgCN\downarrow + H^+$, $K_{eq} = 10^6$

0.01M 0.01M

Q.

$[H^+] = 10^{-2}$ M

$$S = \sqrt{K_{sp}\left(1 + \frac{[H^+]}{K_a}\right)}$$

$$S = \sqrt{10^{-16}\left(1 + \frac{10^{-2}}{10^{-10}}\right)} = \sqrt{10^{-16} \times 10^8}$$

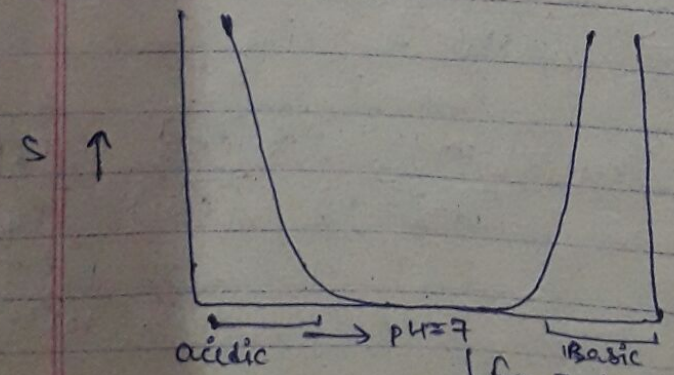
$$\boxed{S = 10^{-4}\ M}$$

(*) Amphoterism :-

Some sparingly soluble compounds (hydroxides) like $Zn(OH)_2$, $Al(OH)_3$, $Cr(OH)_3$ etc are amphoteric and have high solubility in acid solution as well as base solution.

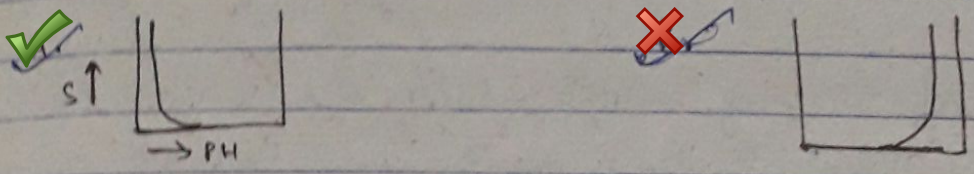
eg. $Zn(OH)_2$ $\xrightarrow{HCl(aq)}$ $ZnCl_2(aq) + H_2O(l)$

$Zn(OH)_2$ $\xrightarrow{NaOH(aq)}$ $Na_2ZnO_2(aq) + H_2O$



(for $Zn(OH)_2$ or $Al(OH)_3$)

Q. which correctly represents solubility of $Fe(OH)_3$.



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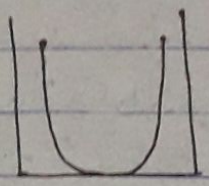

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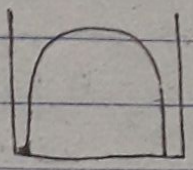

$Fe(OH)_3$ ⟶ Basic Hydroxide.

* Mass balance & Charge Balance :

eg. $H_2C_2O_4 + H_2O(l)$

Mass balance equation (POAC on 'C')

$$[H_2C_2O_4]_0 = [H_2C_2O_4]_t + [HC_2O_4^-]_t + [C_2O_4^{2-}]_t \quad \text{— (i)}$$

Charge balance equation :

Total +ve charge = Total -ve charge

$$1 \times [H^+] = 1 \times [OH^-] + 1 \times [HC_2O_4^-] + 2 \times [C_2O_4^{2-}] \quad \text{— (ii)}$$

eg $Na_3PO_4 \xrightarrow{+H_2O(l)}$

Mass balance : $[Na_3PO_4]_0 = [PO_4^{3-}]_t + [HPO_4^{2-}]_t + [H_2PO_4^-]_t + [H_3PO_4]_t$ — (i)

charge balance :

$$1 \times [H^+] + 1 \times [Na^+] = 1 \times [OH^-] + 3 \times [PO_4^{3-}] + 2 \times [HPO_4^{2-}] + 1 \times [H_2PO_4^-]$$

* Important Questions on Equilibrium (Ionic)
(For IIT-JEE Ad.)

1. Find pH of resulting solution when equal volumes of 0.2M Na_3PO_4 (aq) and 0.3M H_3PO_4 (aq) are mixed.
$K_1 = 10^{-4}$, $K_2 = 10^{-8}$, $K_3 = 10^{-11}$ for H_3PO_4.

(2)

Sol^n:

$$Na_3PO_4 + H_3PO_4 \rightleftharpoons NaH_2PO_4 + Na_2HPO_4$$

(L.R) $V \times 0.2$ | $V \times 0.3$ | $0.2V$ | $0.2V$ mol
-0.2
$= 0.1 \times V$

$$H_3PO_4 + Na_2HPO_4 \longrightarrow 2NaH_2PO_4\,(aq)$$

$0.1 \times V$ (L.R) | $0.2 \times V$ | $0.2V$
$-0.1V$
$= 0.1V$

$$[Na_2HPO_4]_f \text{ (conjugate Base)} = \frac{0.1V}{2V} = 0.05\ M$$

$$[NaH_2PO_4]_f \text{ (Acid)} = \frac{0.2V + 0.2V}{2V}$$
$$= 0.2\ M$$

Conjugate acid - base pair.

$$\underbrace{H_3PO_4}\ |\ \underbrace{H_2PO_4^-}\ |\ \underbrace{HPO_4^{2-}}\ |\ \underbrace{PO_4^{3-}}$$

Buffer, $pH = pK_2 + \log \frac{0.05}{0.20}$
$= 8 + \log 1/4$
$= 8 - 0.6 = 7.4$ Ans.

→ only conjugate acid - base pair remain in solution in appreciably high conc. (for species / derivatives of a polyacidic base or polybasic acid)

(2) Two buffer solutions A and B of pH 4 and 6 respectively are prepared from acid HA and salt NaA. Both the buffers are 0.5 M in HA. Find pH of solution prepared by mixing equal volumes of two buffers.
[pK_a (HA) = 5,], [log 5.05 = 0.7]

Sol^n:

A (pH = 4)

HA + NaA

0.5M C_1

$pH = pK_a + \log \frac{S}{a}$

$4 = 5 + \log \frac{C_1}{0.5}$

$C_1 = 0.05$ M

B (pH = 6)

HA + NaA

0.5M C_2

$pH = 6 = 5 + \log \frac{C_2}{0.5}$

$C_2 = 5$ M

Both buffers A and B are mixed

$[HA] = \frac{V \times 0.5 + V \times 0.5}{2V} = 0.5$ M

$[NaA] = \frac{V \times 0.05 + V \times 5}{2V} = \frac{5.05}{2}$ M

$pH = pK_a + \log \frac{S}{a}$

$= 5 + \log \frac{5.05}{2 \times 0.5}$

$= 5 + 0.7$

$pH = 5.7$ Ans.

(3). 10 ml of 0.1 M H_3A is titrated with 0.1 M NaOH solution. What the ratio of $\frac{[H_3A]}{[A^{3-}]}$ at 2nd equivalent point. Given $K_1 = 7.5 \times 10^{-4}$

$K_2 = 10^{-8}$

$K_3 = 10^{-12}$, for H_3A.

A. Approx 10^{-4}
B. Approx 10^{4}
C. Approx 10^{-7}
D. Approx 10^{+7}

Sol^n:

$$H_3A + 2NaOH(aq) \longrightarrow \underset{(aq)}{Na_2HA} + 2H_2O \quad [2^{nd} \text{ eq. point}]$$

10 ml, 0.1 M 20 ml, 0.1 M 1 mmol

1 mmol 2 mmol

$Na_2HA \longrightarrow$ Salt with amphiprotic anion,

$$[H^+] = \sqrt{K_{a_2} \times K_{a_3}} = \sqrt{10^{-8} \times 10^{-12}}$$

$$[H^+] = 10^{-10} M.$$

$$\left(\frac{[H_3A]}{[A^{3-}]}\right)_{\text{2nd eq. point}} = \frac{[H_3A]}{[A^{3-}]} \times \frac{[H_2A^-]}{[H_2A^-]} \times \frac{[HA^{2-}]}{[HA^{2-}]} \times \frac{[H^+][H^+][H^+]}{[H^+][H^+][H^+]}$$

$$= \frac{1}{K_1} \times \frac{1}{K_2} \times \frac{1}{K_3} \times [H^+]^3$$

$$= \frac{1}{7.5 \times 10^{-4}} \times \frac{1}{10^{-8}} \times \frac{1}{10^{-12}} \times (10^{-10})^3$$

$= \frac{10^{-10}}{7.5} \times 10^{4}$ M

$\approx 10^{-7}$ M.

(4.) The Salt $Zn(OH)_2$ is involved in the following two equilibrium.

$$Zn(OH)_2(s) \rightleftharpoons Zn^{2+}(aq) + 2OH^-(aq)$$
$$K_{sp} = 1.2 \times 10^{-17}$$

$$Zn(OH)_2 + 2OH^-(aq) \rightleftharpoons [Zn(OH)_4]^{2-}(aq), \quad K_f = 0.12$$

Find pH at which solubility of $Zn(OH)_2$ is minimum.

Sol^n:

$$\underset{S}{Zn(OH)_2(s)} \rightleftharpoons \underset{S}{Zn^{2+}} + \underset{S}{2OH^-}, \quad K_{sp} = [Zn^{2+}][OH^-]^2$$
$$1.2 \times 10^{-17} = [Zn^{2+}][OH^-]^2$$

$$Zn(OH)_2 + 2OH^- \rightleftharpoons [Zn(OH)_4]^{2-}(aq), \quad 0.12 = \frac{[Zn(OH)_4]^{2-}}{[OH^-]^2}$$

Applying mass balance, **POAC on Zn**

$$S = [Zn^{2+}] + [Zn(OH)_4]^{2-}(aq)$$

$$S = \frac{1.2 \times 10^{-17}}{[OH^-]^2} + 0.12 \times [OH^-]^2$$

$\frac{d(S)}{d\,pH \text{ or } d\,OH^-} = 0$, for minima.

$$\frac{dS}{d(OH^-)} = \frac{-2 \times 1.2 \times 10^{-17}}{[OH^-]^3} + 0.12 \times 2[OH^-] = 0$$

$$[OH^-]^4 = \frac{1.2 \times 10^{-17}}{0.12} = 10^{-16}$$

$$[OH^-] = 10^{-4}\ M$$

$$[H^+] = 10^{-10}\ M$$

pH of solution at which solubility of $Zn(OH)_2$ is minimum is

= 10 Ans..